



MVJ COLLEGE OF ENGINEERING BANGALORE-560067

(An Autonomous Institute)

MINI PROJECT REPORT

ON

“AI-Powered Portable Health Screening Kiosk for Smart Rural Healthcare Monitoring”

Submitted in practical fulfillment for the award of degree of

MASTER OF TECHNOLOGY

IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

SUBMITTED BY

SPANDANA INDIRA R

1MJ25SAD07

UNDER THE GUIDANCE OF

Prof. P Bindu Madhavi

Assistant Professor

Department of Information Science & Engineering



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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

CERTIFICATE

Certified that the project work titled '*AI-Powered Portable Health Screening Kiosk for Smart Rural Healthcare Monitoring*' is carried out by **SPANDANA INDIRA R(1MJ25SAD07)**, who is bonafide student of MVJ College of Engineering, Bengaluru, in partial fulfilment for the award of Degree of **Master of Technology in Artificial Intelligence and Data Science** of the Visvesvaraya Technological University, Belagavi during the year 2025-2026. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the project report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect of project work prescribed by the institution for the said Degree.

Signature of Guide

Prof.P Bindu Madhavi

Assistant Professor

Department of ISE MVJCE

Bangalore-560067

Signature of HOD

Prof. Shagina P.K.

Head of the Department

Department of ISE MVJCE

Bangalore-560067

External Viva

Name of Examiners

Signature with Date

- 1.
- 2.



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(An Autonomous Institution Affiliated to VTU, Belagavi)

**DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA
SCIENCE**

DECLARATION

I, SPANDANA INDIRA R (1MJ25SAD07) student of second semester MTech, Department of Artificial Intelligence and Data Science, MVJ College of Engineering, Bengaluru, hereby declare that the project titled '*AI-Powered Portable Health Screening Kiosk for Smart Rural Healthcare Monitoring*' has been carried out by me and submitted in partial fulfilment for the award of Degree of **Master of Technology in Artificial Intelligence and Data Science** during the year 2025-26.

Further I declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University.

I also declare that any Intellectual Property Rights generated out of this project carried out at MVJCE will be the property of MVJ College of Engineering, Bengaluru and I will be one of the authors of the same.

Place: Bengaluru

Date:

NAME

USN

SIGNATURE

SPANDANA INDIRA R

1MJ25SAD07

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ABSTRACT

Healthcare accessibility remains a significant challenge in rural and underserved communities due to the shortage of medical professionals, limited healthcare infrastructure, language barriers, and low health awareness. This project presents an AI-Powered Multilingual Healthcare Kiosk, an intelligent and user-friendly system designed to provide preliminary health assessment and disease prediction through an accessible self-service platform.

The proposed solution integrates Google Gemini 2.0 Flash for multilingual symptom understanding and conversational interaction, XGBoost for accurate disease prediction, and SHAP to provide transparent and interpretable predictions that enhance user trust and decision-making. The kiosk enables patients to register, describe their symptoms using either voice or text in English, Hindi, or Kannada, and receive instant health assessments. Speech-to-text processing and multilingual natural language understanding allow users with varying literacy levels to interact effortlessly with the system.

Based on the collected symptoms and patient information, the machine learning model predicts the most probable disease along with its confidence score, possible causes, recommended precautions, suggested medications, and guidance on whether immediate medical consultation is required. SHAP-based explanations clearly highlight the symptoms that contributed to each prediction.

In addition to disease prediction, the system provides automatic PDF medical report generation, secure patient data management using SQLite, longitudinal health record tracking, interactive dashboards for patients and government health administrators, and an AI-powered chatbot for personalized health guidance. The application is implemented using the Flask framework, ensuring a lightweight, scalable, and cost-effective deployment suitable for rural healthcare environments. By combining advanced artificial intelligence with an intuitive healthcare platform, the proposed kiosk bridges the gap between rural communities and quality healthcare services, enabling early disease detection, promoting informed healthcare decisions, and supporting digital healthcare initiatives through an affordable and scalable solution.

ACRONYMS

Acronym	Abbreviation
USN	University Seat Number
SEM	Semester
AI	Artificial Intelligence
GUI	Graphical User Interface
ML	Machine Learning
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
CRUD	Create, Read, Update, Delete
API	Application Programming Interface
CSS	Cascading Style Sheets
DBMS	Database Management System
ER	Entity-Relationship
GUI	Graphical User Interface
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IDE	Integrated Development Environment
IoT	Internet of Things
JSON	JavaScript Object Notation
JWT	JSON Web Token
ML	Machine Learning
MVC	Model-View-Controller
OS	Operating System
REST	Representational State Transfer
RMSE	Root Mean Square Error

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CHAPTER 1 INTRODUCTION

1.1 AIM

The primary aim of this project is to develop an AI-Powered Multilingual Healthcare Kiosk that provides intelligent and accessible preliminary healthcare services using advanced Artificial Intelligence techniques. The system integrates Google Gemini AI for symptom analysis, XGBoost and Ensemble for disease prediction, and SHAP Explainable AI to provide transparent and understandable prediction results.

It enables users to interact through voice and text in multiple languages, making the system suitable for people from different linguistic backgrounds. The kiosk generates personalized health reports, securely stores patient records, and provides real-time healthcare analytics. The proposed system aims to improve early disease screening, reduce the burden on healthcare facilities, and enhance healthcare accessibility, particularly in rural and underserved areas.

Overall, the project seeks to deliver an affordable, reliable, and user-friendly AI-assisted healthcare solution while supporting healthcare professionals in making informed decisions.

The rural healthcare crisis stems from multiple interconnected factors. Healthcare facilities are often located at considerable distances from villages, requiring travel that is both time-consuming and expensive for rural families. Technological advancements in artificial intelligence, particularly in natural language processing and machine learning, offer unprecedented opportunities to address these challenges.

Large Language Models such as Google's Gemini demonstrate remarkable proficiency in understanding and processing medical information across linguistic boundaries. When combined with robust classification algorithms like XGBoost for disease prediction and interpretability frameworks like SHAP, these technologies can create healthcare screening systems that are simultaneously accessible, accurate, and trustworthy.

1.2 MOTIVATION

Healthcare accessibility remains a major challenge, especially in rural areas where hospitals, doctors, and diagnostic facilities are limited. Many patients experience delays in receiving medical attention due to long travel distances, financial constraints, and language barriers.

Although AI has significantly improved disease prediction, most existing healthcare systems lack multilingual support, explainable predictions, and easy-to-use interfaces. These limitations inspired the development of an intelligent healthcare kiosk that can provide quick and reliable preliminary health assessments.

By integrating Gemini AI, XGBoost, Ensemble and SHAP Explainable AI, the proposed system offers accurate disease prediction with transparent explanations. Features such as multilingual interaction, voice assistance, automated report generation, and patient record management make the system more accessible and practical. This project is motivated by the vision of improving healthcare delivery through AI while promoting affordable, inclusive, and technology-driven healthcare services.

Access to quality healthcare remains a persistent challenge in rural regions of developing countries, with India presenting a particularly stark example. Approximately 65% of India's population resides in rural areas, yet these communities have access to merely 30% of the nation's healthcare infrastructure and personnel.

This fundamental imbalance creates a cascade of negative health outcomes, including delayed diagnoses, inadequate treatment interventions, and elevated mortality rates from preventable conditions.

1.3 PROBLEM STATEMENT

India presents a stark healthcare paradox where approximately 65% of its rural population has access to merely 30% of the nation's healthcare infrastructure, creating a fundamental disparity that results in delayed diagnoses, inadequate treatment, and elevated mortality from preventable conditions. Rural villages lack basic healthcare infrastructure with Primary Health Centers frequently located 20 to 50 kilometers away, making geographic isolation an insurmountable barrier for elderly individuals, pregnant women, and persons with mobility limitations.

The shortage of medical professionals is equally severe with approximately one doctor for every 3,000 rural residents compared to one for every 1,500 urban residents, and specialists being virtually absent in rural settings. Language and literacy barriers further compound these challenges as health information is predominantly available in English, excluding approximately 90% of the rural population who communicate primarily in regional languages, preventing effective patient-provider communication and limiting health literacy and treatment adherence.

The absence of regular health screenings leads to disease detection at advanced stages, with rural patients presenting with late-stage diseases 40-60% more frequently than urban counterparts, resulting in three times higher treatment costs and significantly reduced survival rates. Patient health records in rural settings are frequently incomplete, inconsistent, or entirely non-existent, leading to fragmented care, medication errors, and the inability to conduct effective population-level health monitoring for evidence-based policy formulation.

Health literacy surveys indicate that only 15-20% of rural populations possess adequate health literacy to recognize early disease symptoms or understand when professional medical attention is warranted, contributing to delayed healthcare seeking and preventable mortality. Rural households bear a disproportionate financial burden with approximately 70% of healthcare spending being out-of-pocket, pushing families into poverty and creating a vicious cycle where illness leads to poverty and poverty prevents access to care. Despite India's digital transformation, rural healthcare settings remain disconnected from technological advancements.

1.4 EXISTING SYSTEM

Existing healthcare systems primarily rely on hospital visits, telemedicine platforms, and AI-based disease prediction applications to provide medical assistance. Most AI healthcare applications use machine learning algorithms such as Decision Trees, Random Forest, Support Vector Machines (SVM), or Neural Networks to predict diseases based on patient symptoms or medical data.

Although these systems improve diagnostic accuracy, they have several limitations. Most existing solutions support only a single language and lack multilingual voice interaction, making them difficult for rural and non-English-speaking users. In addition, many AI models function as black-box systems, providing predictions without explaining the reasons behind them, which reduces user trust.

Existing systems also have limited features for patient record management, report generation, and healthcare analytics. Furthermore, they often require continuous internet connectivity and are not specifically designed for self-service healthcare kiosks in remote areas. These limitations highlight the need for a more intelligent, transparent, multilingual, and user-friendly healthcare solution.

Major limitations are:

- Limited multilingual support and poor accessibility for regional language users.
- Most AI models do not provide explainable prediction results.
- Lack of voice-enabled interaction for illiterate or elderly users.
- Limited integration with patient record management and PDF report generation.
- No centralized dashboard for healthcare monitoring and analytics.
- Dependence on hospitals or healthcare professionals for initial consultation.
- Not specifically designed for rural and remote healthcare environments.

1.5 PROPOSED SYSTEM

The proposed system is an AI-Powered Multilingual Healthcare Kiosk designed to provide intelligent, accessible, and transparent preliminary healthcare services, particularly for rural and underserved communities. The system combines Google Gemini AI, XGBoost, and SHAP Explainable AI to deliver accurate disease prediction and meaningful explanations for every prediction. Patients can interact with the kiosk through both voice and text in multiple languages such as English, Hindi, and Kannada, making it easy for users with different linguistic backgrounds and literacy levels.

Gemini AI analyzes the patient's symptoms and extracts relevant medical information, while the XGBoost model predicts the most probable disease based on the processed features. SHAP Explainable AI identifies the key symptoms that influence the prediction, improving transparency and user confidence.

The system securely stores patient information in an SQLite database and automatically generates a detailed PDF health report containing symptoms, predicted disease, confidence score, precautions, and recommended specialist consultation. The report can also be shared through the WhatsApp Cloud API for easy access and future reference. Additionally, the system provides a real-time analytics dashboard for monitoring patient records, disease trends, and healthcare statistics.

The proposed healthcare kiosk is designed to support early disease screening, reduce unnecessary hospital visits, improve healthcare accessibility, and assist healthcare professionals in making timely and informed decisions. It serves as an intelligent decision-support system rather than replacing professional medical diagnosis, thereby contributing to a more efficient, affordable, and technology-driven healthcare ecosystem.

Main objectives are:

- a) Development of a Multilingual, Voice-Enabled Healthcare Kiosk Interface.
- b) Integration of AI-Powered Symptom Analysis and Disease Prediction
- c) Implementation of Explainable AI for Transparent Decision-Making
- d) Development of Automated Health Documentation and Report Generation
- e) Implementation of Real-Time Analytics Dashboards.
- f) Development of Rule-Based Fallback for System Reliability.
- g) Generate a comprehensive text-based summary report detailing traffic analysis results contribution to Rural Healthcare Accessibility.
- h) Creation of a scalable, affordable healthcare screening solution deployable in rural Primary Health Centers.

CHAPTER 2

LITERATURE SURVEY

This is the Research paper [1] by Maleki Varnosfaderani and Forouzanfar (2024) on "The Role of AI in Hospitals and Clinics: Transforming Healthcare in the 21st Century" critically assesses AI's integration across healthcare domains including clinical decision-making, hospital operations, medical image analysis, and patient care through AI-powered wearables. The methodology involves a comprehensive review with case studies examining how AI has transformed specific healthcare domains and discussing challenges and possible solutions. A key advantage is the exploration of methodologies for assessing AI healthcare solutions and the importance of data privacy and bias mitigation. However, the paper highlights remaining challenges including ethical deployment concerns and the need for responsible technology use. This research emphasizes the importance of interdisciplinary dialogue between researchers, clinicians, and technologists for AI implementation.

This is the Research paper [2] by a systematic review (2024) on "Artificial Intelligence in Rural Healthcare Delivery: Bridging Gaps and Enhancing Equity through Innovation" systematically reviewed 109 studies published between 2019 and 2024 to investigate AI's transformative potential in addressing rural healthcare challenges. The methodology followed PRISMA guidelines and Covidence software for screening articles from PubMed, Embase, Web of Science, IEEE Xplore, and Scopus databases. A key advantage is the identification of significant promise for AI applications such as predictive analytics, telemedicine platforms, and automated diagnostic tools in improving healthcare accessibility, quality, and efficiency. However, barriers including infrastructural limitations, data quality concerns, and ethical considerations remain. The study highlights that Multimodal Foundation Models and Large Language Models offer particularly transformative potential for rural healthcare by augmenting human capacity, reducing diagnostic delays, and democratizing access to expertise.

Research Paper [3] by Quanita et al. (2024) on "Precision Death Forecasting of Dengue Outbreaks in Bangladesh by Blending Statistical and Machine Learning Models" leverages a comprehensive dataset from Bangladesh spanning 2019 to 2023 to improve dengue outbreak predictions. The methodology employs SARIMA, XGBoost, and Random Forest Regression models on 900 daily records capturing temporal and climatic variables. A key advantage is that XGBoost achieved superior predictive accuracy with the lowest MSE of 2.6108 and RMSE of 1.6158, markedly surpassing other models. However, the SARIMA model showed limitations in managing non-linear trends. This research underscores the significance of integrating machine learning techniques with time series analysis for enhancing the precision of health predictions, demonstrating XGBoost's versatility beyond clinical diagnosis to public health surveillance applications. The study further highlights the potential of ML models to serve as early warning systems for epidemic-prone diseases, enabling proactive public health interventions and resource allocation. Additionally, the research emphasizes the importance of incorporating climate and environmental data into predictive models.

Research Paper [4] by a study (2023) on "An Interpretable Approach with Explainable AI for the Detection of Cardiovascular Disease" proposes an interpretable approach using XAI to detect cardiovascular disease by combining different machine learning algorithms with XAI techniques. The methodology employs SHAP to provide human-understandable explanations of model predictions, emphasizing both accuracy and interpretability. A key advantage is achieving competitive performance metrics where ANN achieved the highest accuracy of 91%, while solving the black-box problem associated with AI models. However, the study's focus on cardiovascular disease alone may limit broader applicability. This research highlights the importance of XAI to bridge the gap between ML models and clinical decision-making, fostering trust in AI-driven healthcare solutions. The study demonstrates that SHAP explanations enable clinicians to validate model predictions by identifying specific features contributing to each diagnosis, thereby enhancing clinical acceptance and facilitating integration into routine practice. Furthermore, the research emphasizes that explainability not only builds trust but also helps in identifying potential biases in model predictions.

This is the Research paper [5] by Abbasi et al. (2023) on "AI in Remote Observation and m-Telemedicine" analyzes the current uses, advantages, and disadvantages of AI in remote observation and mobile telemedicine. The methodology involves a review article examining AI applications for healthcare delivery in underserved areas. A key advantage is the identification of AI's potential to enhance remote healthcare delivery and monitoring. However, challenges including infrastructural limitations and data quality concerns remain. This research provides a comprehensive overview of AI applications in remote healthcare settings.

This is the Research paper [6] by Esteva et al. (2022) on "A Guide to Deep Learning in Healthcare" provides a comprehensive overview of deep learning applications in healthcare including medical imaging, disease diagnosis, drug discovery, and personalized medicine . The methodology involves a review article examining how deep learning models have demonstrated remarkable performance across various medical domains. A key advantage is the demonstration that AI systems have achieved diagnostic accuracy comparable to or exceeding human experts in several medical domains. However, challenges including data requirements, interpretability, and regulatory approval remain significant barriers. This research provides a foundational framework for understanding AI applications in healthcare, emphasizing the importance of human-centered AI design and transparent decision-making, which are core principles of the proposed healthcare kiosk .

This is the Research paper [7] by Kumar et al. (2022) on "Prediction of Skin Disease with Improved Accuracy through Comparative Analysis of Support Vector Machine and XGBoost" compares SVM and XGBoost models for skin disease prediction . The methodology involves preprocessing a Kaggle dataset of clinical and demographic variables, training and testing both algorithms, and conducting an independent samples t-test to evaluate statistical significance. A key advantage is the finding that XGBoost achieved 94% accuracy, significantly outperforming SVM which achieved only 59.6% accuracy with a statistically significant difference ($p < 0.001$). However, the study's reliance on a single dataset and the need for extensive data collection for accurate skin disease prediction models present limitations. This research demonstrates XGBoost's superior performance in medical classification tasks, reinforcing its selection as the primary prediction model for the healthcare kiosk.

This is the Research paper [8] by Quanita et al. (2020) on "Precision Death Forecasting of Dengue Outbreaks in Bangladesh by Blending Statistical and Machine Learning Models" leverages a comprehensive dataset from Bangladesh spanning 2019 to 2023 to improve dengue outbreak predictions. The methodology employs SARIMA, XGBoost, and Random Forest Regression models on 900 daily records capturing temporal and climatic variables. A key advantage is that XGBoost achieved superior predictive accuracy with the lowest MSE of 2.6108 and RMSE of 1.6158, markedly surpassing other models. However, the SARIMA model showed limitations in managing non-linear trends. This research underscores the significance of integrating machine learning techniques with time series analysis for enhancing the precision of health predictions, demonstrating XGBoost's versatility beyond clinical diagnosis to public health surveillance applications.

This is the Research paper [9] by Sirigineedi et al. (2019) on "Symptom-Based Disease Prediction: A Machine Learning Approach" focuses on leveraging machine learning algorithms for disease prediction based on symptoms using a dataset comprising 132 symptoms and 41 diseases. The methodology involves data preprocessing, feature selection techniques, and exploration of various algorithms including decision trees, support vector machines, random forests, and XGBoost. A key advantage is the use of cross-validation and hyperparameter tuning to enhance model performance, with XGBoost emerging as one of the top-performing models due to its capacity to manage complicated relationships within the data. However, the authors acknowledge that the model needs further validation on diverse datasets and regular updates to remain relevant in clinical settings. This research provides a robust methodological foundation for symptom-based disease prediction, directly supporting the core diagnostic functionality of the healthcare kiosk.

This is the Research paper [10] by Rajkomar, Dean, and Kohane (2019) on "Machine Learning in Medicine" published in the New England Journal of Medicine, is a highly cited and foundational review that systematically examines how machine learning is poised to transform nearly every facet of healthcare delivery.

The authors, leading experts in biomedical informatics, meticulously outline the key areas where ML can have the greatest impact, including enhancing diagnostic accuracy, personalizing treatment planning, predicting patient prognoses, and optimizing clinical workflows to expand the reach of expert knowledge .

The methodology is a comprehensive synthesis of the existing literature, designed to demystify ML for a clinical audience by providing clear illustrative examples of its applications and explaining how these algorithms function. A core advantage of this work is its balanced perspective: while it powerfully advocates for ML's potential to improve clinical decision-making and reduce errors, it rigorously catalogues the major challenges and vulnerabilities that must be addressed . This includes critical issues like the quality and bias of training data, the "black box" problem of interpretability, and the significant implementation barriers in real-world clinical settings.

Crucially, the paper concludes with a sobering and essential argument that ML systems are designed to complement, not replace, human clinical expertise. It emphasizes that medicine will always require human judgment, empathy, and nuanced understanding—elements that algorithms cannot replicate.

This research provides essential, enduring insights for integrating ML into clinical workflows, fundamentally framing AI as a powerful decision-support tool that must be deployed with caution, robust validation, and a clear understanding of its limitations to ensure safe and effective patient care.

CHAPTER 3

SYSTEM REQUIREMENT AND SPECIFICATION

3.1 HARDWARE REQUIREMENTS:

- RAM : Minimum 8 GB (16 GB Recommended)
- Processor : Intel Core i5 (10th Generation or above) / AMD Ryzen 5 or higher
- Display : 15.6-inch Monitor / Touch Screen Display (Recommended)
- Storage : 256 GB SSD or above
- Input Devices : Keyboard, Mouse, Touch Screen
- Audio Devices : Microphone and Speakers
- **Internet Connection:** Required for installing dependencies and using the gTTS service.

For the Processor (CPU), runs multiple concurrent processes including the Flask web server, SQLite database operations, XGBoost model loading, SHAP explanation generation, and real-time voice processing. The Gemini AI API integration requires substantial memory for handling symptom analysis requests, while the machine learning models and their dependencies consume significant RAM during prediction and explanation generation. With 8 GB RAM, the system can handle basic operations, but 16 GB is recommended to ensure smooth multitasking, faster response times, and improved user experience, particularly when processing multiple patient consultations simultaneously.

High computational power for executing machine learning predictions, generating SHAP explanations, processing voice inputs through speech recognition, and rendering the user interface. XGBoost and SHAP operations are computationally intensive and require fast processing to deliver real-time results. The 10th Generation Intel Core i5 or AMD Ryzen 5 processors provide the necessary performance while maintaining power efficiency, making them suitable for deployment in rural healthcare settings where power availability may be limited.

A 15.6-inch display offers adequate screen space to present the multilingual user interface, symptom input forms, health reports, and analytics dashboards clearly. The touch screen capability is recommended to eliminate the dependency on traditional input devices, making the interface more intuitive for users with limited digital literacy. In rural healthcare environments, touch interaction simplifies navigation and reduces the learning curve, ensuring elderly patients and individuals with disabilities can easily access the system.

The system requires storage for the operating system, Python environment, all project code files, machine learning models (XGBoost and SHAP), SQLite database, generated PDF health reports, QR code images, and system configuration files. A Solid State Drive (SSD) provides faster read and write speeds compared to traditional hard drives, which is critical for rapid model loading, database operations, and report generation. The enhanced reliability and durability of SSDs make them ideal for deployment in rural areas with limited maintenance support.

These input devices are essential for patient registration, symptom entry, and system navigation. The keyboard facilitates typing patient details and symptoms, while the mouse enables easy navigation through the interface. The touch screen provides an alternative, more intuitive input method for users with limited computer literacy. This combination ensures accessibility for diverse user groups, including those with physical limitations.

The microphone enables speech-to-text functionality, allowing patients to describe symptoms verbally, which is particularly beneficial for illiterate users. The speakers support text-to-speech output, delivering health information and recommendations audibly. These audio devices are critical for the system's multilingual voice interface, ensuring effective communication with patients regardless of their literacy level.

An active internet connection is necessary for installing all dependencies and for integrating with cloud-based services such as Gemini AI for symptom analysis and gTTS for text-to-speech conversion. Additionally, WhatsApp Cloud API integration for report delivery requires internet connectivity. However, the system incorporates a rule-based fallback engine to ensure basic functionality even in offline scenarios, providing reliability in areas with limited connectivity.

3.2 SOFTWARE REQUIREMENTS:

- Operating System: Windows10/11
- Deep Learning Framework : Hugging Face Transformer
- Programming Language : Python 3.11 or above
- Backend Framework : Flask
- Frontend Technologies : HTML5, CSS3, JavaScript, Bootstrap 5
- Database : SQLite
- AI Model : Google Gemini AI
- Machine Learning Algorithm : XGBoost, Ensemble
- Explainable AI : SHAP (SHapley Additive Explanations)
- Development Environment : Visual Studio Code

The healthcare kiosk system is developed and tested on Windows 10 and 11 operating systems, which provide a stable and user-friendly environment for deployment. Windows offers broad compatibility with the required Python libraries, Flask framework, and machine learning tools such as XGBoost, SHAP, and Hugging Face Transformers. It also provides seamless integration with hardware components such as touch screens, microphones, and speakers, making it suitable for deployment in rural healthcare settings where Windows-based systems are commonly available and easier to maintain.

The operating system's built-in accessibility features, such as speech recognition and text-to-speech, complement the kiosk's voice-enabled interface, enhancing usability for patients with varying abilities. Windows Defender, ensuring that patient health data remains protected against unauthorized access. The system's compatibility with Windows Update ensures that the kiosk can receive regular security patches and performance improvements, maintaining system reliability over extended deployment periods.

Hugging Face Transformers is used as the deep learning framework to access and utilize state-of-the-art pre-trained models for natural language understanding tasks. The framework provides easy access to the Gemini AI model through its unified API, enabling efficient symptom analysis and multilingual processing across English, Hindi, and Kannada languages. Hugging Face Transformers simplifies the integration of complex transformer-based models, reducing development time while ensuring high performance and accuracy. The framework's extensive model hub offers a wide range of pre-trained models that can be fine-tuned for specific healthcare applications, providing flexibility for future enhancements.

Additionally, the framework supports the integration of the rule-based fallback engine and facilitates model fine-tuning and deployment, ensuring the system can adapt to evolving healthcare requirements. The framework's efficient tokenization and batch processing capabilities enable rapid processing of patient symptom descriptions, ensuring minimal wait times for users.

Python is the primary programming language for developing the healthcare kiosk system due to its extensive libraries and frameworks for machine learning, web development, and data processing. The XGBoost algorithm, SHAP explainability, and Flask backend are all implemented in Python, ensuring seamless integration and efficient development. Python 3.11 or above is recommended for enhanced performance, improved error handling, and better support for asynchronous programming, which is essential for handling concurrent API requests and real-time voice processing.

The language's simplicity and readability enable rapid development and easy maintenance, while its extensive community support provides access to a wealth of resources and troubleshooting assistance. Python's interoperability with C/C++ libraries ensures optimal performance for computationally intensive tasks such as XGBoost prediction and SHAP explanation generation, while its robust testing frameworks enable comprehensive validation of system components.

Flask is a lightweight and flexible micro-framework for building web applications in Python. It serves as the backend framework for handling HTTP requests, routing, and API integration. Flask provides the foundation for integrating the Gemini AI, XGBoost model, and SHAP explainer, while also managing database operations and session handling. Its simplicity and modularity enable rapid development and easy deployment, with built-in development server and debugger facilitating testing and troubleshooting. Flask's extensibility through numerous extensions allows for seamless integration of additional features such as user authentication, file uploads, and email notifications as the system evolves. The framework's Jinja2 templating engine enables dynamic HTML rendering, supporting the multilingual interface and personalized patient dashboards. Flask's support for RESTful API design ensures seamless communication between the frontend and backend, while its handling of asynchronous tasks enables efficient processing of voice input and report generation without blocking the user interface.

The frontend of the healthcare kiosk is built using HTML5 for structuring content, CSS3 for styling and responsive design, and JavaScript for interactivity and dynamic content updates. Bootstrap 5 is utilized as the frontend framework to ensure responsive layouts, consistent styling, and accessibility across different devices and screen sizes. These technologies collectively create an intuitive, multilingual, and user-friendly interface that adapts to various display sizes, from large kiosk touchscreens to smaller tablets and smartphones. HTML5 provides semantic structure and multimedia support, enabling voice input through the Web Speech API and dynamic content loading. CSS3 enables visually appealing designs with smooth animations and transitions, enhancing user engagement. JavaScript powers the core interactivity, including symptom input validation, real-time voice recording, and dynamic result updates, ensuring a responsive and engaging patient experience. Bootstrap 5 provides pre-built components such as cards, modals, and navigation bars, accelerating development while maintaining visual consistency and accessibility standards.

SQLite is used as the lightweight relational database management system for the healthcare kiosk. It provides a simple, zero-configuration storage solution for managing patient records, consultation histories, and language preferences. SQLite's portability and small footprint make it ideal for deployment in rural areas where resource availability may be limited. The database supports efficient querying and indexing, enabling fast retrieval of

patient history and health records. SQLite's ACID compliance ensures data integrity, preventing data corruption in case of system failures. The database stores patient registration details including name, phone, gender, and age, as well as consultation records containing symptoms, predicted diseases, confidence scores, and SHAP explanations. The language preferences table enables personalized user experiences across sessions. SQLite's ability to handle concurrent read operations makes it suitable for the kiosk's multi-user environment.

Google Gemini AI is integrated as the primary AI engine for natural language understanding and symptom analysis. Gemini processes patient symptom descriptions in multiple languages, extracting structured symptom data and generating disease predictions. The multilingual capabilities of Gemini enable effective communication with patients in English, Hindi, and Kannada, ensuring accessibility across diverse populations. Gemini's advanced reasoning capabilities allow it to understand complex symptom descriptions and generate appropriate follow-up questions, enhancing diagnostic accuracy. The model's ability to process both text and voice inputs through a unified interface simplifies integration and improves user experience. Gemini's cloud-based architecture ensures access to the latest model versions without requiring local updates, while its API provides flexible integration options and scalability for handling varying patient loads. The model's continuous learning capabilities enable adaptation to new symptom patterns and evolving healthcare knowledge, ensuring the system remains current and effective.

XGBoost is the primary machine learning algorithm for disease prediction, selected for its high accuracy, interpretability, and efficiency in handling structured datasets. The algorithm employs gradient boosting with regularization techniques to reduce overfitting and improve generalization. XGBoost's ability to handle missing data and its built-in cross-validation capabilities ensure robust performance across diverse patient populations. The ensemble approach combines multiple models to enhance prediction accuracy and robustness, ensuring reliable and consistent performance across different disease categories. The ensemble model integrates XGBoost with rule-based systems and additional ML algorithms, providing multiple layers of prediction and fallback mechanisms. This ensemble approach improves overall system reliability and ensures the system remains functional even when individual components experience issues. XGBoost's efficient implementation supports rapid training and inference, enabling real-time disease

predictions that are essential for timely healthcare interventions.

SHAP is implemented to provide interpretability for disease predictions by explaining the contribution of each symptom to the model's decision. SHAP generates both visual and text explanations in patient-friendly language, building trust and enabling clinicians to validate predictions, thereby fostering confidence in the AI-driven healthcare system. The technique's foundation in game theory ensures consistent and locally accurate explanations, making it suitable for healthcare applications where understanding and trust are paramount. SHAP's ability to handle complex models like XGBoost and its model-agnostic nature make it a versatile tool for enhancing transparency across the system's ensemble approach. SHAP explanations highlight key symptoms influencing predictions, enabling patients to understand why a specific disease was predicted, while also helping healthcare providers identify potential biases or errors. The visualizations are designed to be accessible to both patients and healthcare professionals, with clear color coding and intuitive representations.

Visual Studio Code is used as the development environment for building the healthcare kiosk system. It provides robust support for Python development, debugging, version control, and extension management. Visual Studio Code's integrated terminal, code completion, and error highlighting streamline the development process, ensuring efficient implementation and maintenance of the system. The IDE's extensive extension marketplace provides tools for Flask development, SQLite database management, and API testing, enhancing productivity. Visual Studio Code's integrated Git support enables efficient version control and collaboration, while its debugging capabilities allow for rapid identification and resolution of issues. The environment's support for multiple programming languages facilitates seamless frontend and backend development, ensuring consistency across the entire system. Visual Studio Code's remote development capabilities enable deployment testing and troubleshooting, ensuring smooth operation across different environments.

3.3 FUNCTIONAL REQUIREMENTS:

- **Patient Registration :** The patient registration functionality enables users to create a secure profile within the healthcare kiosk system by entering essential personal details including their full name, age, gender, phone number, and residential address. The system employs robust validation mechanisms to verify the accuracy and completeness of the entered information, checking for valid phone number formats, appropriate age ranges, and complete address details before any data is stored in the database. Once validated, the patient information is securely stored in the SQLite database with encryption measures to protect sensitive personal data, ensuring that all registration records are maintained for future consultations and longitudinal health record management. This functionality forms the foundation of the system's patient-centric approach, enabling personalized healthcare delivery and continuity of care across multiple visits.
- **Multilingual Language Selection :** The multilingual language selection feature empowers users to choose their preferred communication language from English, Hindi, or Kannada, ensuring the healthcare kiosk accommodates diverse linguistic backgrounds. Upon language selection, the system dynamically adapts all interface elements including instructions, menus, prompts, error messages, and notifications to the chosen language, providing a seamless and culturally appropriate user experience. The voice input and voice output capabilities are also synchronized with the selected language, enabling patients to speak their symptoms and receive spoken health guidance in their native tongue. This functionality is particularly critical in rural healthcare settings where patients may have limited proficiency in English, ensuring that language barriers do not impede access to quality healthcare screening.
- **Symptom Entry and Analysis :** The symptom entry and analysis module serves as the primary patient interaction point, allowing users to describe their health concerns through intuitive text input or voice-enabled speech recognition. Patients can type their symptoms using the on-screen keyboard or speak naturally using the integrated microphone, with the system capturing and

processing the input in real-time. The Google Gemini AI engine analyzes the symptom description, extracting structured medical information including symptom types, severity indicators, duration, and affected body systems. When the initial symptom description lacks sufficient detail for accurate prediction, the system intelligently generates follow-up questions to gather additional information, progressively refining the symptom profile and improving prediction accuracy through this interactive dialogue.

- **Disease Prediction :** The disease prediction engine leverages the XGBoost machine learning algorithm to generate accurate and reliable disease predictions based on the processed symptom data. The system converts the extracted symptoms and associated medical information into structured feature vectors, which are then fed into the trained XGBoost model to predict the most probable disease condition. The prediction results are presented to the user with a clear confidence score indicating the model's certainty in the diagnosis, enabling patients to understand the reliability of the assessment. The entire prediction process is optimized for speed, delivering results within a reasonable response time to ensure that patients receive timely health insights without unnecessary delays, facilitating early intervention and reducing healthcare-seeking anxiety.
- **Explainable AI (SHAP) :** The Explainable AI component implements SHAP (SHapley Additive exPlanations) to provide transparent and interpretable explanations for every disease prediction made by the system. This functionality analyzes the contribution of each symptom to the final prediction, identifying which symptoms most significantly influenced the diagnostic decision and quantifying their relative importance. The explanations are presented in a simple, user-friendly format that includes both visual representations such as bar charts showing symptom contributions and clear textual descriptions explaining why the system predicted a particular disease. By making the AI's decision-making process transparent and understandable, this functionality builds trust among patients and healthcare providers, enabling informed decision-making and facilitating validation of the system's predictions.

- **Health Report Generation :** The health report generation functionality automatically produces comprehensive PDF reports immediately following each consultation, creating a permanent record of the patient's health assessment. Each report includes patient demographic details, documented symptoms, predicted disease, confidence score, severity assessment, precautionary measures, recommended medications, and specialist consultation guidance. The generated reports are professionally formatted with the healthcare kiosk branding and include a QR code for easy verification and digital access. Users can conveniently download the report to their device or print a physical copy, enabling them to share their health information with family members, community health workers, or medical professionals for follow-up care, thereby enhancing care coordination and patient empowerment.
- **Patient Record Management :** The patient record management system provides secure and efficient storage and retrieval of all patient-related data through the SQLite database. This functionality maintains comprehensive health records including patient registration information, complete consultation history, symptom documentation, prediction results, SHAP explanations, and generated health reports. Administrators and authorized personnel can search for patient records using various criteria such as name, phone number, or consultation date, and update records as needed while maintaining data integrity and audit trails. The system ensures that previous consultation reports and health history remain readily accessible for future reference, enabling healthcare providers to track disease progression, monitor treatment effectiveness, and make informed decisions based on longitudinal patient data.
- **Dashboard and Analytics:** The dashboard and analytics module provides an interactive, real-time visualization interface for monitoring healthcare trends and system performance. This functionality displays comprehensive patient statistics including total registrations, consultation frequency, disease distribution patterns, and demographic breakdowns through intuitive graphical representations such as bar charts, pie charts, and trend lines. Healthcare administrators can analyze disease prevalence across different regions,

identify emerging health patterns, and monitor changes in patient demographics over time, with all dashboard components updating automatically as new patient data is added to the system. This functionality supports evidence-based healthcare decision-making, enabling administrators to allocate resources effectively, detect potential disease outbreaks early, and formulate targeted public health interventions based on real-time data insights.

3.4 NON-FUNCTIONAL REQUIREMENTS:

- **Performance:** Process video frames and generate annotated output at a minimum of Frames Per Second (FPS) for near real-time analysis.
- **Maintainability:** Ensure the codebase is well-structured, modular, and comprehensively documented for ease of understanding, debugging, and future updates.
- **Scalability:** Handle processing of video files smoothly and adapting to future increases in video data volume.
- **Usability:** Provide a clear and straightforward setup process, with easily configurable parameters and intuitive presentation of analytical results.
- **Security:** Safeguard processed video data, analytical reports, and system configurations against unauthorized access or tampering.
- **Interoperability:** Be compatible with common video file formats (e.g., MP4, AVI), for diverse types video inputs and data source integration.

CHAPTER 4

SYSTEM DESIGN

The proposed AI-Powered Multilingual Healthcare Kiosk follows a modular three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer to ensure efficient communication, scalability, and maintainability. The Presentation Layer serves as the user interface where patients interact with the kiosk through a touch-enabled web interface using voice or text input in their preferred language from English, Hindi, or Kannada.

This layer incorporates the Web Speech API for voice-based symptom description and health guidance, ensuring accessibility for users with varying literacy levels and accommodating elderly patients and individuals with disabilities through high contrast ratios, large touch targets, and clear visual hierarchy. The Application Layer acts as the core processing unit, handling patient registration, multilingual language processing, symptom analysis through Google Gemini AI using Natural Language Processing, disease prediction through the XGBoost machine learning model, explainable AI through SHAP for transparent decision-making, automated PDF report generation, WhatsApp Cloud API integration for report delivery, and real-time analytics dashboards for healthcare monitoring.

The Gemini AI processes patient symptom descriptions, extracts structured medical information, and intelligently generates follow-up questions when needed to enhance prediction accuracy. The extracted features are transformed into feature vectors and passed to the XGBoost model, which predicts the most probable disease along with a confidence score. The SHAP module identifies the key symptoms influencing each prediction and generates user-friendly explanations, building trust and enabling patients to understand the reasoning behind their health assessment. The Data Layer manages all data storage through an SQLite database with SQLAlchemy ORM, maintaining patient details, consultation history, language preferences, and generated reports, while also managing file storage for PDF reports.

The processed results are securely stored, providing patients with convenient access to their health information.

4.1 SYSTEM ARCHITECTURE

The system architecture is divided into four primary layers: the Presentation Layer, the Application Layer, the AI Layer, and the Data Layer. The Presentation Layer handles all user interactions through the web interface, providing multilingual support and both voice and text input capabilities. The Application Layer manages business logic, patient registration, session management, and workflow coordination. The AI Layer encompasses all artificial intelligence and machine learning components including Gemini AI for symptom analysis, XGBoost for disease prediction, and SHAP for model explainability.

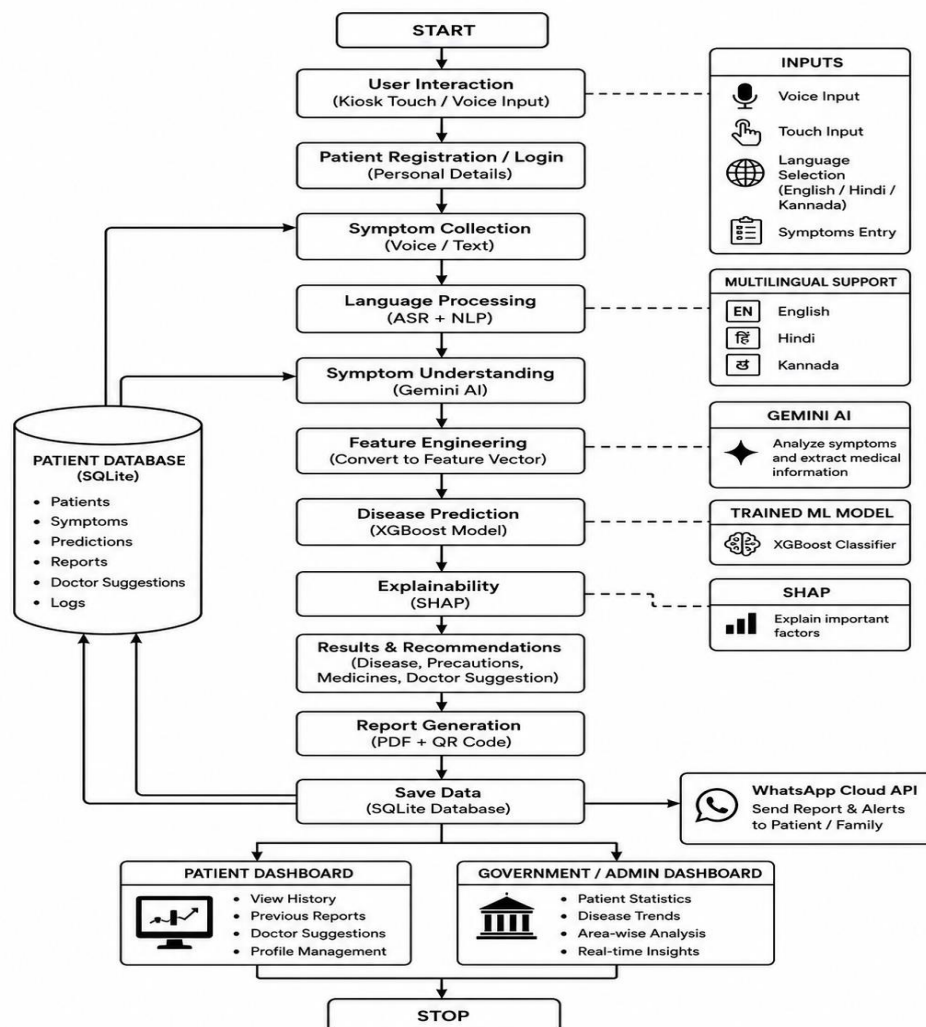


Figure 4.1 System Architecture

The Presentation Layer forms the user interface of the healthcare kiosk, developed using HTML, CSS, and JavaScript to provide an interactive, responsive, and accessible experience for patients and administrators. The Patient Interface includes registration forms where patients enter demographic information including name, age, gender, and mobile number, with real-time validation to ensure data accuracy. The multilingual interface allows patients to select their preferred language from options including English, Hindi, Tamil, Telugu, Kannada, Malayalam, Bengali, and Marathi, with all prompts, instructions, and responses displayed in the selected language. The symptom entry interface provides both voice input through Web Speech API and text input capabilities, enabling patients with different abilities and preferences to describe their symptoms effectively.

Voice input is particularly beneficial for patients with limited literacy, visual impairments, or those who find typing difficult, as it allows natural spoken descriptions of symptoms. The results display interface presents disease predictions with confidence scores, probability distributions across disease categories, and SHAP explanations with feature importance rankings and influence directions in an understandable format. The report viewing interface allows patients to access their comprehensive health reports in PDF format directly through the web interface. The Administrator Interface provides dashboard access for healthcare administrators to view patient statistics, disease trends, area-wise analysis, and real-time insights. The dashboard includes interactive visualizations such as bar charts for disease distribution, line charts for consultation trends, KPI cards for key metrics, and summary tables for recent consultations, all designed to facilitate data-driven decision-making for healthcare resource allocation and public health monitoring.

The Application Layer is the core business logic tier of the healthcare kiosk, responsible for orchestrating the complete workflow from patient registration to report delivery. The Patient Registration Module validates patient information, checks for existing patients using mobile number as unique identifier, generates a unique patient ID using timestamp-based formatting with the pattern "PATYYYYMMDDHHMMSS", and stores the information in the SQLite Database. The registration process ensures that each patient is properly identified and all consultations are associated with the correct patient record. The Language Selection Module manages the multilingual interface, ensuring that all system prompts, symptom analysis requests, prediction results, SHAP explanations, and generated reports are presented in the patient's preferred language. The language preference is stored in the patient's profile and applied consistently across all interactions.

The Symptom Collection Module captures patient symptoms through voice input using Web Speech API for speech-to-text conversion or text input, validates that meaningful symptoms have been entered, and prepares the data for AI analysis. The Session Management Module maintains patient session state

AI-Powered Portable Health Screening Kiosk for Smart Rural Healthcare Monitoring

throughout the consultation process, ensuring that data flows correctly between modules and that the patient's progress is tracked from registration through report generation. The Workflow Coordination Module orchestrates the sequence of operations, ensuring that symptom analysis is completed before feature engineering, feature engineering before disease prediction, disease prediction before explainability, and all steps completed before report generation and delivery. The Report Delivery Module manages the distribution of health reports through multiple channels, including web interface display, PDF download, and WhatsApp delivery, providing patients with flexible access options.

The AI/ML Layer is the intelligent processing engine of the healthcare kiosk, comprising three major components: the Symptom Understanding Module using Gemini AI, the Disease Prediction Module using XGBoost, and the Explainability Module using SHAP. The Symptom Understanding Module receives patient symptom descriptions in multiple languages and processes them through Google's Gemini AI, which has been specifically prompted to extract structured medical information from free-text descriptions. Gemini AI analyzes the symptom description and extracts symptoms list, severity levels categorized as mild/moderate/severe, duration of symptoms, frequency of occurrence, affected body parts, and additional medical notes. This structured analysis transforms the patient's natural language description into organized medical data that can be processed by the prediction engine, ensuring that critical medical information is captured accurately and consistently regardless of how the patient described their symptoms. The module constructs carefully designed prompts that guide Gemini AI to extract specific medical information in a consistent JSON format, enabling reliable parsing and feature extraction.

The Feature Engineering Module converts the structured symptom analysis into numerical feature vectors that the XGBoost machine learning model can process. Features include symptom count, average severity score calculated using severity mapping mild=1, moderate=2, severe=3, duration weight based on time units hours=1, days=2, weeks=3, months=4, years=5, frequency weight based on occurrence patterns occasional=1, frequent=2, constant=3, and patient age. These engineered features are designed to capture the most clinically relevant aspects of the symptom presentation, providing the XGBoost model with comprehensive input for accurate disease classification.

The Disease Prediction Module uses the trained XGBoost classifier, which has been optimized with hyperparameters including 100 estimators, maximum depth of 6, learning rate of 0.1, subsample ratio of 0.8, and column sample ratio of 0.8 to prevent overfitting and ensure generalization. The XGBoost model processes the feature vectors and generates probability distributions across eight disease categories including Common Cold, Influenza, Gastroenteritis, Malaria, Tuberculosis, Asthma, Allergy, and Dengue. The model's gradient boosting capabilities ensure high prediction accuracy by sequentially improving model performance through iterative learning, making it particularly effective for healthcare classification

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tasks where accuracy is critical. The Confidence Scoring sub-module identifies the disease with maximum probability and calculates the confidence score as the maximum probability multiplied by 100, providing patients with a clear indication of the prediction reliability.

The Explainability Module uses SHAP (SHapley Additive exPlanations) to compute feature importance values for each symptom and patient characteristic, providing transparent and understandable explanations for the disease prediction. The SHAP TreeExplainer processes the feature vector and calculates SHAP values for each feature, determining which symptoms most influenced the prediction and whether each feature contributed positively increasing disease likelihood or negatively decreasing disease likelihood. The module ranks features by importance, generates explanations in patient-friendly language, and prepares the explanations for display in the web interface and inclusion in the health report. This explainability component is critical for building trust in the AI system, enabling validation of predictions by healthcare professionals, and providing educational value by helping patients understand the relationship between their symptoms and potential diseases. By combining accurate prediction with transparent explanation, the AI/ML Layer ensures that the healthcare kiosk provides not only reliable disease assessments but also understandable reasoning behind those assessments.

The Data Layer manages all data storage and retrieval operations using SQLite Database, providing secure and efficient storage for patient information, consultation records, prediction results, generated reports, and system logs. The Patient Database stores patient demographic information including patient ID, name, age, gender, mobile number, preferred language, and registration date. The database supports patient lookup by mobile number for duplicate checking and patient record retrieval for consultations, ensuring that each patient's medical history is properly maintained.

The Consultation Database stores consultation records including patient ID, symptom descriptions, structured symptom analysis, predicted disease, confidence score, probability distribution, prediction date, report path, and WhatsApp delivery status. This comprehensive storage enables historical retrieval, trend analysis, and longitudinal health monitoring for patients. The Report Database stores generated health reports with their associated filenames, file paths, generation timestamps, and consultation IDs, supporting report retrieval and management. The System Logs Database stores system operation logs, user activities, error records, and performance metrics, supporting system monitoring, debugging, and performance optimization. The database schema includes proper indexing on frequently queried fields such as patient ID and mobile number to ensure efficient retrieval, and all data access operations are encapsulated in data access objects that provide a clean interface for the application layer, promoting maintainability and data consistency.

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The system integrates with two primary external services: the WhatsApp Cloud API and Google's Gemini AI API. The WhatsApp Cloud API integration enables the healthcare kiosk to deliver health reports directly to patients' mobile devices. The system constructs personalized WhatsApp messages with patient names, attaches the PDF health report as a document, and sends it through the WhatsApp Business API to the patient's registered mobile number. The integration includes delivery status tracking, confirming successful report delivery to patients and logging any delivery failures for follow-up. This seamless delivery mechanism ensures that patients receive immediate access to their health reports without requiring additional downloads or portal logins, making healthcare information conveniently accessible on their mobile devices. The Google Gemini AI API integration enables intelligent symptom analysis by processing patient symptom descriptions and extracting structured medical information.

The system constructs prompts that guide Gemini AI to extract specific medical information including symptoms, severity levels, duration, frequency, body parts, and additional notes, and the API returns structured JSON output that is parsed and processed by the system. The integration supports multilingual symptom analysis, allowing patients to describe symptoms in their preferred language and receive structured medical information consistently across languages.

The system architecture incorporates multiple security and privacy measures to protect patient health information. Patient data is transmitted securely using HTTPS protocol, ensuring that all communication between the web interface and the backend server is encrypted. Patient registration information, consultation records, and health reports are stored in the SQLite Database with appropriate access controls, ensuring that only authorized users patients and administrators can access specific data. Patient authentication is managed through session-based authentication, ensuring that patient sessions are properly controlled and expired after periods of inactivity. Personal health information is not shared with external services except through secure API integrations Gemini AI and WhatsApp Cloud API that comply with data protection regulations. The architecture also includes audit logging to track all data access and modification operations, supporting compliance with healthcare data protection requirements.

The proposed system is deployed as a web-based application using the Flask framework, running on a Python server environment. The application is containerized for easy deployment and scalability. The architecture supports deployment on various platforms including on-premises servers, cloud infrastructure, or containerized environments such as Docker, providing flexibility for different healthcare deployment scenarios. The system is designed to handle multiple concurrent patient consultations efficiently, with proper load management and resource allocation. Backup and recovery mechanisms ensure that patient data is protected against hardware failures and data corruption, with regular automated backups of the SQLite Database ensuring data recoverability.

The modular architecture of the proposed system supports future enhancements and scalability. The AI/ML Layer can be extended with additional models for specialized disease prediction or for analyzing additional medical parameters. The multilingual support can be extended to include additional languages as required by different regions. The WhatsApp integration can be enhanced with additional notification types and interactive features. Future enhancements may include integration with Electronic Health Records EHR systems, telemedicine capabilities for real-time consultation with healthcare professionals, wearable device integration for continuous health monitoring, and advanced predictive analytics for identifying population health trends.

The architecture supports cloud deployment for improved scalability and availability, with the ability to handle increasing numbers of patients and consultations as the system grows. Integration with healthcare information systems and interoperability standards such as FHIR Fast Healthcare Interoperability Resources would enable seamless data exchange with existing healthcare infrastructure. Machine learning model retraining pipelines can be implemented to continuously improve prediction accuracy as more data becomes available. The architecture is also designed to support real-time healthcare provider alerts for severe predictions, enabling immediate escalation to healthcare professionals when urgent conditions are detected.

4.2 DATA FLOW DIAGRAM

The Level-0 Data Flow Diagram (DFD), also known as the Context Diagram, provides a high-level overview of the proposed AI-Powered Multilingual Healthcare Kiosk System. It illustrates the overall interaction between the external entities and the central healthcare kiosk without showing the internal processing modules. The primary objective of the Level-0 DFD is to represent how data enters the system, how it is processed, and how the final outputs are delivered to different stakeholders.

This diagram serves as the foundation for understanding the complete system architecture and establishes the boundary of the proposed healthcare kiosk by identifying all external entities that interact with the system and the data flows that occur between them.

In the proposed system, the AI-Powered Healthcare Kiosk System acts as the central process that communicates with four external entities: Patient, Administrator, SQLite Database, and WhatsApp Cloud API. The patient initiates the consultation by entering registration details such as name, age, gender, and contact information. After successful registration, the patient selects the preferred language from multiple options including English, Hindi and Kannada, and

provides symptoms through either voice input or text input. These details are transmitted to the healthcare kiosk for further processing. The patient also receives disease predictions, SHAP explanations, health reports, and WhatsApp notifications from the system, completing the bidirectional communication between the patient and the healthcare kiosk.

- DFD Level 0

The healthcare kiosk processes the patient's information using Artificial Intelligence and Machine Learning techniques to predict the most probable disease. The system analyzes symptoms using Google's Gemini AI, which extracts structured medical information including symptoms list, severity levels, duration, frequency, affected body parts, and additional medical notes. The XGBoost machine learning model processes engineered features including symptom count, average severity, duration weight, frequency weight, and patient age to generate disease predictions with confidence scores and probability distributions across multiple disease categories. After prediction, the SHAP explainer computes feature importance values, providing transparent and understandable explanations for the disease prediction. The system generates a detailed health report containing the patient's details, identified symptoms, predicted disease, confidence score, SHAP explanations, precautionary measures, and healthcare recommendations.

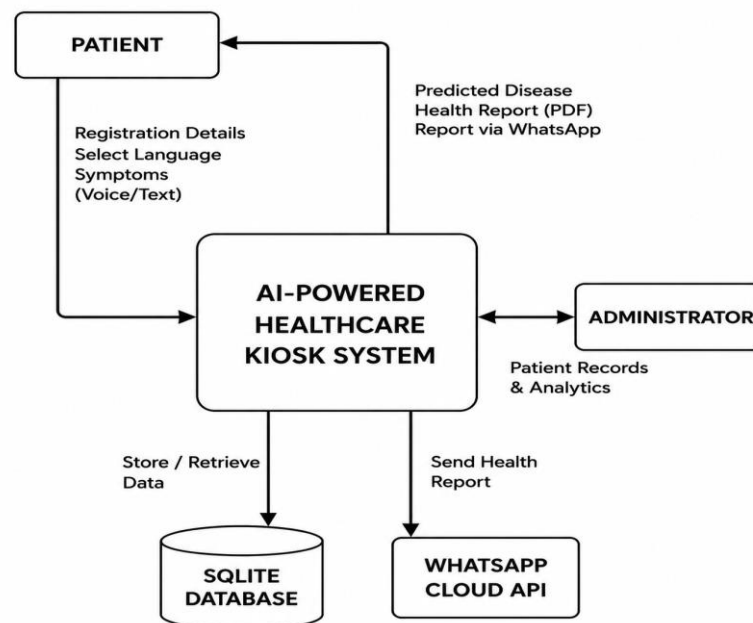


Figure 4.2.1: Data Flow Diagram Level 0

The generated report is displayed to the patient and is also shared through the WhatsApp Cloud API, allowing patients to access their medical reports conveniently on their mobile devices.

The healthcare kiosk communicates continuously with the SQLite Database, where all patient information, consultation records, disease predictions, SHAP explanations, generated reports,

and system logs are securely stored. The database supports retrieval of previous consultation records whenever required, ensuring proper maintenance of patient history and enabling efficient healthcare management. The database also stores the trained XGBoost model parameters and maintains user authentication information for secure access to the system. This continuous data exchange ensures that all healthcare interactions are properly documented and organized for future reference and analysis.

The Administrator interacts with the healthcare kiosk to monitor the overall functioning of the system. The administrator can access patient records, review consultation history, monitor disease statistics, analyze healthcare trends, and generate analytical reports through the administrative dashboard. This functionality assists healthcare authorities in monitoring disease prevalence, identifying regional health patterns, understanding system utilization, and making informed decisions regarding healthcare resource allocation. The administrator also manages user accounts, monitors system performance, and ensures the overall reliability and security of the healthcare kiosk.

The WhatsApp Cloud API serves as an external entity that enables the healthcare kiosk to deliver health reports directly to patients' mobile devices. The system sends personalized WhatsApp messages containing the generated PDF health report as a document attachment to the patient's registered mobile number. This seamless delivery mechanism ensures that patients receive immediate access to their health reports without requiring additional downloads or portal logins. The WhatsApp Cloud API also supports delivery status tracking, allowing the system to confirm successful report delivery to patients.

Overall, the Level-0 Data Flow Diagram presents a simplified representation of the complete healthcare kiosk by illustrating the major external entities and the flow of information between them. It demonstrates that the patient provides input to the system, the healthcare kiosk performs intelligent disease prediction and report generation, the SQLite database manages secure data storage and retrieval, the WhatsApp Cloud API delivers digital health reports, and the administrator utilizes stored information for monitoring and healthcare analytics. This high-level representation serves as the foundation for understanding the detailed workflows described in the subsequent Level-1 and Level-2 Data Flow Diagrams.

- DFD Level 1

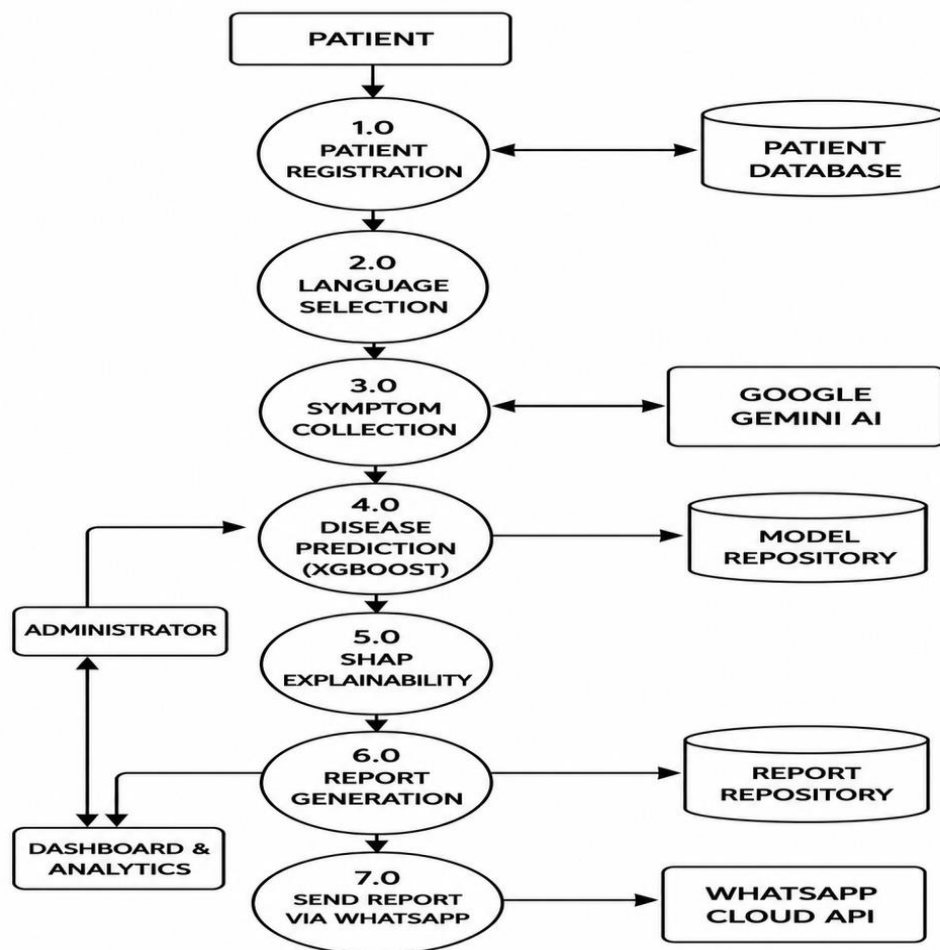


Figure 4.2.2 DFD Level 1

The proposed system follows a sequential, modular pipeline architecture for AI-driven preliminary healthcare diagnostics. The data flow is divided into two primary segments: the patient-facing interaction flow and the backend infrastructure.

The patient journey begins with the registration module, which serves as the entry point of the system. A unique patient identifier is generated, and demographic metadata such as age, gender, and medical history is captured and stored as structured attributes, ensuring session persistence and traceability for subsequent operations. Following registration, the localization layer is activated through the language selection module, which determines the natural language processing context for the entire session. The selected language parameter dictates the prompt engineering strategy for the large language model and ensures that all generated reports are output in a language comprehensible to the patient.

The primary data acquisition occurs in the symptom collection module, which employs an interactive conversational agent powered by the Google Gemini AI API. This agent conducts dynamic, adaptive questioning using a symptom ontology, and patient responses are parsed, normalized, and transformed into a structured feature vector compatible with the downstream machine learning model. This module effectively bridges unstructured patient narratives to structured clinical features. The structured feature vector is then fed into a pre-trained XGBoost classifier, which is a gradient-boosted decision tree ensemble known for handling non-linear feature interactions and providing high predictive accuracy with minimal overfitting. The model outputs a probability distribution over a predefined set of disease classes and returns the top-k probable diagnoses, with the model artifact being fetched from the model repository.

To ensure model interpretability—a critical requirement for clinical acceptance—the system applies SHAP explainability. SHAP values are computed locally for each prediction to quantify the marginal contribution of every symptom feature toward the final output. This generates a feature-importance summary such as force plots or summary bars that explain why a particular disease was predicted, thereby adding transparency and trust to the AI-driven decision-making process. The system then aggregates the patient's demographic data, symptom logs, prediction results, and SHAP explanation visualizations into a structured report, which is formatted into a standardized, human-readable document such as PDF or HTML. This report is subsequently stored in the report repository for future reference and audit trails.

The final delivery layer invokes the WhatsApp Cloud API, Meta's official business API, to asynchronously transmit the generated report to the patient's registered WhatsApp number. This module handles message templating, media attachment, and delivery status callbacks, ensuring real-time, secure, and convenient report dissemination.

The backend infrastructure comprises four key components that support the entire pipeline. The Google Gemini AI serves as the natural language processing engine for the symptom collection module, processing user queries, generating follow-up questions, and extracting structured symptom data via few-shot prompting and JSON mode output constraints. The model repository is a version-controlled storage system such as MLflow or S3 that hosts the serialized XGBoost model artifacts, feature encoders, and scalers, from which the prediction module fetches the latest

production-ready model at runtime. The report repository is a document storage system such as MongoDB GridFS or cloud object storage that persists all generated patient reports, indexed by patient ID and timestamp, enabling historical retrieval and compliance with data retention policies. The WhatsApp Cloud API provides a RESTful interface that acts as the gateway for sending templated messages and media files to end-users, abstracting the underlying telephony infrastructure.

In summary, the system follows a microservices-inspired modular design where each processing stage is decoupled and interacts through well-defined data schemas. Data flows unidirectionally from registration to delivery, with intermediate persistence points for model artifacts and generated reports. The integration of XGBoost enables high-performance classification, SHAP enforces explainability, and the combination of Gemini AI with WhatsApp API ensures a human-centric and accessible user interface. This architecture is scalable, auditable, and clinically explainable, making it suitable for deployment in real-world telemedicine and preliminary screening applications.

4.3 FLOW CHART

The Flow Chart is an essential design tool that graphically represents the sequence of operations performed by the proposed AI-Powered Multilingual Healthcare Kiosk. Unlike the Data Flow Diagram (DFD), which focuses on the movement of data between different modules, the flow chart illustrates the logical execution of processes, decision points, and the overall workflow of the application from the beginning to the end. It provides a clear understanding of how the system processes patient registrations, performs symptom analysis, predicts diseases using XGBoost, provides SHAP explainability, generates health reports, and delivers reports through WhatsApp integration. The flow chart presents the complete operational sequence of the proposed healthcare kiosk, making it easier for developers, researchers, and users to understand how different modules interact with one another. It also highlights important decision points such as patient registration validation, symptom entry verification, prediction confidence assessment, and report generation confirmation.

The operational workflow of the proposed system proceeds through the following detailed steps:

- **Start**

This symbol represents the beginning of the execution of the AI-Powered Multilingual Healthcare Kiosk. The application is initialized, and all required software components including Flask web framework, database connection, Gemini AI model, XGBoost prediction model, and SHAP explainer are loaded and configured for operation.

- **Patient Registration**

- I. The patient opens the healthcare kiosk web interface on their device.
- II. The patient enters demographic information including name, age, gender, and mobile number.
- III. The system validates all input fields and checks for existing patients using mobile number as unique identifier.
- IV. The system generates a unique patient ID using timestamp-based formatting and stores the information in the database.

- **Registration Successful?**

- I. This is an important decision point that verifies whether the patient has successfully registered.
- II. If NO: The system displays validation errors and requests the patient to correct the entered information.
- III. If YES: The patient is successfully registered and proceeds to the symptom entry stage.

- **Select Preferred Language**

- I. After successful registration, the patient selects their preferred language from multilingual options.
- II. Available languages include English, Hindi, Kannada
- III. The system updates the interface language and ensures all prompts, symptom analysis, predictions, explanations, and reports are displayed in the selected language.

- **Enter Symptoms**

- I. The patient describes their symptoms through voice input using Web Speech API or through text input in their selected language.
- II. The system captures the symptom description and validates that symptoms have been entered.
- III. Voice input capability enables patients with limited literacy or typing abilities to describe their symptoms naturally.

- **Symptoms Entered?**

- I. This decision point determines whether valid symptoms have been entered by the patient.
- II. If NO: The system displays an error message and requests the patient to enter symptoms through voice or text.
- III. If YES: The symptom description proceeds to the analysis stage.

- **Analyze Symptoms Using Gemini AI**

- I. The system constructs structured prompts for Google's Gemini AI model.
- II. Gemini AI processes the symptom description and extracts structured medical information.
- III. The system extracts symptoms list, severity levels categorized as mild/moderate/severe, duration of symptoms, frequency of occurrence, affected body parts, and additional medical notes.
- IV. The structured analysis transforms free-text symptom descriptions into organized medical data.

- **Perform Feature Engineering**

- I. The system converts the structured symptom analysis into numerical feature vectors.
- II. Features include symptom count, average severity score, duration weight based on time units, frequency weight based on occurrence patterns, and patient age.
- III. These engineered features serve as input for the XGBoost prediction model.

- **Predict Disease Using XGBoost**

- I. The XGBoost classifier processes the engineered feature vector.
- II. The system generates probability distributions across multiple disease categories.
- III. Disease categories include Common Cold, Influenza, Gastroenteritis, Malaria, Tuberculosis, Asthma, Allergy, and Dengue.
- IV. The system calculates confidence scores as the maximum probability multiplied by 100 for percentage representation.
- V. The predicted disease is determined as the category with the highest probability.

• Compute SHAP Explanation

- I. The SHAP explainer processes the feature vector and computes feature importance values.
- II. The system determines which symptoms and patient characteristics most influenced the prediction.
- III. Each feature is evaluated for positive or negative contribution to the disease prediction.
- IV. Features are ranked by importance and explanations are generated for patient understanding.

• Display Prediction Results

- I. The system displays the predicted disease name to the patient.
- II. The confidence score is shown as a percentage.
- III. Probability distribution across all disease categories is displayed.
- IV. SHAP explanations with feature importance rankings and influence directions are presented.

• Generate Health Report

- I. The Report Generation Module compiles all health information into a structured PDF report.
- II. The report includes patient details, symptom analysis summary, prediction results with confidence score, SHAP feature importance explanations, and general healthcare recommendations.
- III. The report is formatted professionally using ReportLab library with proper styling.
- IV. The report is saved with a unique filename containing patient ID and timestamp.

• **Store Report in Database**

- I. The generated health report path is stored in the database.
- II. The consultation record is updated with the report path and timestamp.
- III. All patient information, symptom data, prediction results, and report details are preserved for future retrieval and historical tracking.

• **Continue Analysis?**

- I. This decision point checks whether the patient wants to perform another consultation.
- II. If YES: The system returns to the symptom entry stage for a new consultation.
- III. If NO: The application proceeds to terminate the current session.

• **View Dashboard Analytics (Administrator)**

- I. The administrator accesses the healthcare analytics dashboard.
- II. The system displays total patient count, total consultations, and disease distribution patterns.
- III. Average confidence scores and recent consultation activities are shown.
- IV. These analytics help healthcare administrators monitor system performance and public health trends.

• **End**

This symbol represents the successful completion of the healthcare consultation workflow. The patient exits the application after receiving their health assessment, viewing SHAP explanations, accessing the health report. The proposed Flow Chart clearly illustrates the complete operational workflow of the AI-Powered Multilingual Healthcare Kiosk.

It demonstrates how patients securely register, select their preferred language, enter symptoms through voice or text, receive Gemini AI-based symptom analysis, obtain XGBoost-based disease predictions with confidence scores, view SHAP explainability results, generate comprehensive PDF health reports, store reports in the database.. The inclusion of multiple decision points, such as registration verification, symptom validation, WhatsApp delivery confirmation, and continuation of analysis, enhances the

robustness and reliability of the system.

By integrating multilingual support, voice input, AI-powered symptom analysis, machine learning-based disease prediction, explainable AI, the flow chart demonstrates how the proposed healthcare kiosk systematically transforms patient symptoms into meaningful and actionable health insights, making preliminary healthcare assessment more accessible, understandable, and effective for patients from diverse linguistic and demographic backgrounds.

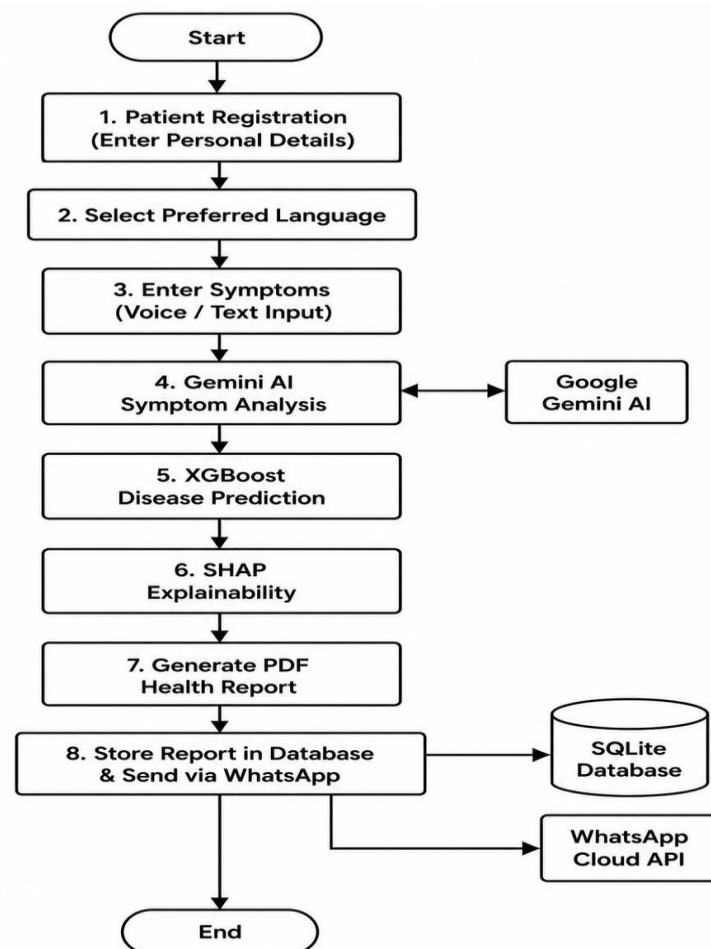


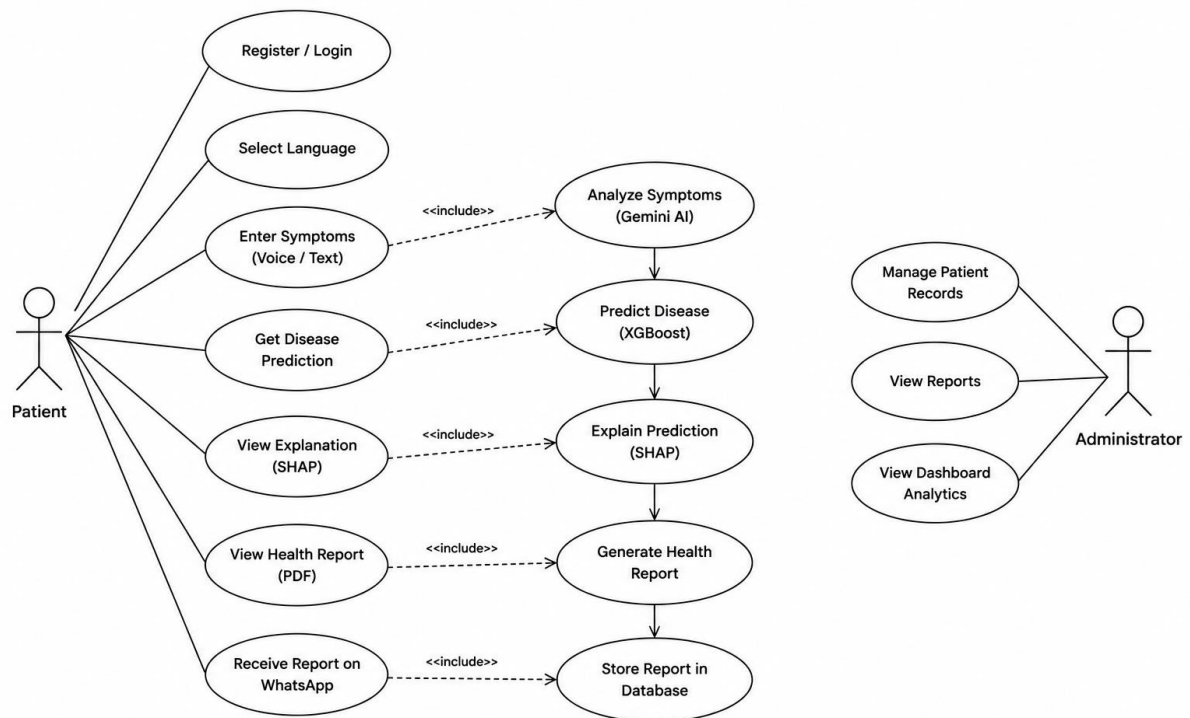
Figure 4.3.1 Flow Chart

Overall, the flow chart provides a comprehensive procedural blueprint that explains how raw retail data is transformed into actionable business intelligence through advanced analytics, visualization, and Large Language Model-based conversational decision support, enabling organizations to make accurate, efficient, and data-driven business decisions.

4.4 USE CASE DIAGRAM

The Use Case Diagram serves as a fundamental tool in visualizing the functional requirements and interactions between users and the proposed "AI-Powered Multilingual Healthcare Kiosk" system. Unlike flow charts that focus on procedural sequences or Data Flow Diagrams that emphasize data movement, a use case diagram graphically illustrates the high-level functionalities of a system from the perspective of its users, known as actors. It captures the system's behavioral requirements by showing the various ways users interact with the system to achieve specific goals.

It captures the system's behavioral requirements by showing the various ways users interact with the system to achieve specific goals. For this project, the use case diagram is indispensable for identifying and documenting the system's core capabilities, clarifying user roles, and establishing the boundaries of the software application. It provides a clear communication bridge between stakeholders, developers, and end-users, ensuring that all functional requirements are understood and addressed. This detailed functional mapping is vital for requirements gathering, system design, testing, and for anyone seeking to understand the scope and capabilities of the software solution.



4.4.1 Use Case Diagram

The Use Case Diagram serves as an effective communication tool among developers, researchers, project evaluators, and end users by presenting the system functionalities in a simple and understandable manner. It also assists during requirement analysis, system design, implementation, testing, and future system enhancements. The above Use Case Diagram clearly illustrates the main actors and the various use cases that define the functional scope of the proposed AI-Powered Multilingual Healthcare Kiosk.

[1] USER

Represented by a stick figure, the USER is the primary actor who directly interacts with the AI-Powered Multilingual Healthcare Kiosk. The user may be a patient seeking preliminary health assessment or any individual requiring healthcare information and disease prediction. The patient performs all major interactions with the system, including registering with demographic information, selecting a preferred language from multiple multilingual options, entering symptoms through voice or text input, receiving disease predictions with confidence scores, viewing SHAP explainability results to understand prediction reasoning, accessing comprehensive health reports in PDF format, and receiving reports directly on WhatsApp for convenient access. Their role represents both the initiation of the healthcare consultation process and the utilization of the generated health insights for informed medical decision-making and follow-up with healthcare professionals.

[2] SYSTEM

The SYSTEM actor represents the AI-Powered Multilingual Healthcare Kiosk itself. It performs all computational tasks automatically after receiving input from the patient. The system validates patient information, generates unique patient IDs, processes symptom descriptions through Google's Gemini AI for structured medical information extraction, performs feature engineering, predicts diseases using XGBoost with confidence scores, computes SHAP feature importance values for model explainability, generates comprehensive PDF health reports using ReportLab, stores patient information and consultation records in the database, delivers reports through WhatsApp Business API integration, and provides healthcare analytics and dashboard statistics for administrators. The SYSTEM actor highlights the intelligent automation capabilities of the proposed healthcare platform without requiring continuous user intervention.

USE CASES

[1] Patient Registration

This use case represents the registration process where the PATIENT enters demographic information including name, age, gender, and mobile number. The system validates all input fields, checks for existing patients using mobile number as unique identifier, generates a unique patient ID using timestamp-based formatting, and stores the information in the database. Only registered patients are allowed to access the healthcare consultation services. Actor: PATIENT

[2] Select Language

This use case allows the PATIENT to select their preferred language from multiple multilingual options including English, Hindi, Tamil, Telugu, Kannada, Malayalam, Bengali, and Marathi. The system updates the interface language and ensures that all prompts, symptom analysis, predictions, explanations, and reports are displayed in the selected language. This feature improves accessibility for patients from diverse linguistic backgrounds. Actor: PATIENT

[3] Enter Symptoms

This use case enables the PATIENT to describe their symptoms through voice input using Web Speech API or through text input in their selected language. The system captures the symptom description, validates that symptoms have been entered, and prepares the data for analysis. Voice input capability enables patients with limited literacy or typing abilities to describe their symptoms naturally. Actor: PATIENT

[4] Analyze Symptoms

Once symptoms are entered, the SYSTEM automatically performs symptom analysis using Google's Gemini AI. The system constructs structured prompts for Gemini AI, processes the symptom description, and extracts structured medical information including symptoms list, severity levels categorized as mild/moderate/severe, duration of symptoms, frequency of occurrence, affected body parts, and additional medical notes. This step transforms free-text symptom descriptions into organized medical data for accurate disease prediction. Actor: SYSTEM

[5] Predict Disease

This use case represents the analytical core of the application. The SYSTEM performs feature engineering by converting symptom analysis into numerical feature vectors including symptom count, average severity, duration weight, frequency weight, and patient age. The XGBoost classifier processes these features and generates disease predictions with confidence scores and probability distributions across multiple disease categories including Common Cold, Influenza,

Gastroenteritis, Malaria, Tuberculosis, Asthma, Allergy, and Dengue. These results form the basis for preliminary health assessment and decision-making. Actor: SYSTEM

[6] Explain Prediction

This use case provides model explainability and interpretability using SHAP (SHapley Additive exPlanations). The SYSTEM computes feature importance values for each symptom and patient characteristic, determines whether each feature positively or negatively influenced the prediction, ranks features by importance, and generates understandable reasoning for the disease prediction. This transparency builds trust in the AI system and helps patients understand the relationship between their symptoms and potential diseases. Actor: SYSTEM

[7] View Disease Prediction

This use case allows the PATIENT to view the predicted disease along with confidence scores and probability distributions across all disease categories. The system displays the most likely disease, confidence percentage, and probability breakdown for each disease, enabling patients to understand their preliminary health assessment results clearly. Actor: PATIENT

[8] View SHAP Explanation

This use case enables the PATIENT to view the SHAP explainability results with feature importance rankings, influence directions showing positive or negative contribution, and understandable reasoning for the disease prediction. Patients can see which symptoms most influenced their prediction and how each symptom contributed to the diagnosis, improving healthcare literacy and understanding. Actor: PATIENT

[9] Generate Health Report

This use case represents the report generation functionality of the system. After completing the disease prediction and SHAP explanation, the SYSTEM prepares comprehensive health reports containing patient details, symptom analysis summaries, prediction results with confidence scores, SHAP feature importance explanations, and general healthcare recommendations. The report is formatted professionally using ReportLab library and saved with a unique filename containing patient ID and timestamp. Actor: SYSTEM

[10] Store Report in Database

Once the health report is generated, the SYSTEM automatically stores the report path in the database and associates it with the consultation record. This ensures that all health reports are properly documented, organized, and available for future retrieval, historical tracking, and follow-up consultations. Actor: SYSTEM

[11] View Health Report

AI-Powered Portable Health Screening Kiosk for Smart Rural Healthcare Monitoring

This use case allows the PATIENT to access and view their comprehensive health report in PDF format directly through the healthcare kiosk interface. The patient can download, print, or share the report with healthcare professionals during follow-up consultations, bridging the gap between preliminary AI-based assessment and professional medical care. Actor: PATIENT

[12] Receive Report on WhatsApp

This use case enables the PATIENT to receive the generated health report as a document attachment on their registered WhatsApp number through the WhatsApp Business API integration. The system constructs a personalized message with patient name, attaches the PDF report, sends it through the WhatsApp API, and tracks delivery status. This seamless delivery mechanism ensures immediate access to health reports without requiring additional downloads or portal logins. Actor: PATIENT

[13] Manage Patient Records

This use case represents the administrative functionality where the ADMINISTRATOR manages patient information, reviews consultation history, updates patient records, and maintains the overall patient database. The administrator can view all registered patients, access their consultation history, and ensure data integrity for effective healthcare service delivery. Actor: ADMINISTRATOR

[14] View Reports

This use case allows the ADMINISTRATOR to view generated health reports for all patients, review consultation details, access historical medical records, and monitor the overall healthcare service delivery. The administrator can filter, search, and retrieve specific reports for quality assurance and healthcare monitoring purposes. Actor: ADMINISTRATOR

[15] View Dashboard Analytics

This use case enables the ADMINISTRATOR to access comprehensive healthcare statistics and analytics dashboard including total patient count, total consultations, disease distribution patterns, average confidence scores, and recent consultation activities. These analytics provide valuable information for healthcare administrators to understand disease prevalence, system utilization, and prediction performance for data-driven decision-making. Actor: ADMINISTRATOR

RELATIONSHIPS

The connecting lines between actors and use cases illustrate the interaction between the PATIENT, ADMINISTRATOR, and the SYSTEM.

[1] PATIENT Relationships

The PATIENT is directly associated with Patient Registration, Select Language, Enter Symptoms, View Disease Prediction, View SHAP Explanation, View Health Report, and Receive Report on WhatsApp. These interactions represent the major activities performed by patients while utilizing the proposed healthcare kiosk for preliminary health assessment and report access.

[2] ADMINISTRATOR Relationships

The ADMINISTRATOR is directly associated with Manage Patient Records, View Reports, and View Dashboard Analytics. These interactions represent the administrative and monitoring activities performed by healthcare administrators to manage the system and track healthcare trends.

[3] SYSTEM Relationships

The SYSTEM is responsible for Analyze Symptoms using Gemini AI, Predict Disease using XGBoost, Explain Prediction using SHAP, Generate Health Report using ReportLab, and Store Report in Database. These use cases represent the intelligent and automated functionalities performed by the proposed healthcare platform after receiving patient input.

In conclusion, the Use Case Diagram provides a concise yet comprehensive representation of the functional capabilities of the proposed AI-Powered Multilingual Healthcare Kiosk. It clearly identifies the PATIENT, ADMINISTRATOR, and the SYSTEM as the primary actors while illustrating essential functionalities such as patient registration, multilingual language selection, symptom entry through voice or text, Gemini AI-based symptom analysis, XGBoost-based disease prediction, SHAP explainability, PDF report generation, WhatsApp integration, database storage, and healthcare analytics. The diagram establishes a clear understanding of the system's functional scope and serves as an effective communication tool during requirement analysis, system design, implementation, and testing. Furthermore, it demonstrates how the proposed platform integrates Artificial Intelligence, machine learning, and multilingual support to simplify preliminary healthcare consultations, enabling patients to obtain meaningful health insights and make informed medical decisions through an intuitive and accessible healthcare interface.

CHAPTER 5

IMPLEMENTATION

The implementation begins with importing essential libraries including Flask for building the web application framework, SQLAlchemy for database management and ORM functionality, Google Gemini AI for intelligent symptom analysis and natural language processing, XGBoost for accurate disease prediction using gradient boosting, SHAP for model explainability and feature importance visualization, ReportLab for professional PDF report generation, and Requests for WhatsApp API integration. The Flask application is configured with a secret key for session management and a SQLite database URI for persistent storage, while CORS is enabled to allow cross-origin requests from frontend applications.

The Gemini AI model is initialized using the API key from environment variables, and the pre-trained XGBoost model along with its SHAP explainer are loaded from pickle files to ensure immediate availability for prediction requests. The database schema defines two primary models: Patient which stores demographic information including unique patient ID generated with timestamps, name, age, gender, mobile number validated for 10 digits, preferred language for multilingual support, and registration date, along with Consultation which stores medical data including symptom descriptions, predicted diseases, confidence scores, prediction timestamps, report file paths, and WhatsApp delivery status. The relationship between patients and consultations is established through foreign key constraints, enabling comprehensive medical history tracking for each patient.

The patient registration API endpoint handles POST requests by validating input data, checking for required fields including name, age, gender, and mobile, validating mobile number format to ensure exactly 10 digits, and checking for existing patients using mobile number as unique identifier. If a patient already exists, the system returns the existing patient ID and a confirmation message, otherwise it generates a new unique patient ID using timestamp-based formatting, creates a new Patient object, saves it to the database, and returns success response with the patient ID. The symptom analysis endpoint processes patient symptom descriptions by constructing structured prompts for Gemini AI that request extraction of specific medical information including symptoms list, severity levels for each symptom categorized as mild, moderate, or severe, duration of symptoms, frequency of occurrence, affected body parts, and additional medical notes.

The Gemini AI model processes these prompts in multiple languages as specified by the patient's preference and returns structured JSON output that is parsed and returned to the frontend for further processing, with proper error handling for invalid inputs or API failures. The feature engineering function converts symptom analysis into numerical feature vectors by computing symptom count, average severity using mapping of mild=1, moderate=2, severe=3, duration weights based on time units where hours=1, days=2, weeks=3, months=4, years=5, frequency weights from occasional=1, frequent=2, constant=3, and patient age. The engineered features are combined into a numpy array with standardized feature names, ensuring consistent input format for the XGBoost model. The prediction endpoint receives the engineered features and patient data, uses the trained XGBoost classifier to predict disease probabilities.

The SHAP explainability endpoint computes feature importance values using the TreeExplainer, determining how each symptom and patient characteristic contributed to the disease prediction, whether the contribution was positive or negative, and ranking features by absolute importance. This explainability component transforms the machine learning model from a black box into an interpretable decision-support tool, providing patients and healthcare providers with clear understanding of why the model made a particular prediction.

The model training script loads symptom-disease datasets, performs train-test split with 80% for training and 20% for testing, trains the XGBoost classifier with optimized hyperparameters including 100 estimators, maximum depth of 6, learning rate of 0.1, subsample ratio of 0.8, and column sample ratio of 0.8 to prevent overfitting. Model performance is evaluated using accuracy, precision, recall, and F1-score metrics with weighted averaging for multi-class classification, and the trained model is saved as a pickle file for production deployment.

The implementation emphasizes modular design with separate functions and route handlers for each major component including registration, symptom analysis, feature engineering, prediction, explainability, report generation, WhatsApp integration, dashboard analytics, and model training. This modular approach improves code maintainability, simplifies debugging and testing, and enables easy updates to individual components without affecting the entire system.

5.1 METHODOLOGY

The development of the proposed AI-Powered Multilingual Healthcare Kiosk follows a systematic and structured methodology that combines Artificial Intelligence, machine learning, natural language processing, multilingual support, and healthcare analytics to simplify medical consultations and preliminary diagnosis. The primary objective of this methodology is to transform patient-reported symptoms into accurate disease predictions through intelligent analysis, enabling users to receive preliminary health assessments, generate medical reports, and access healthcare information through voice and text interactions in multiple languages.

This methodology serves as the foundation for designing, implementing, and integrating various functional modules of the proposed system while ensuring accuracy, reliability, scalability, and user accessibility. The methodology acts as a bridge between the conceptual design and the practical implementation of the proposed healthcare kiosk. It clearly defines the sequence of activities involved in patient registration, multilingual symptom analysis, disease prediction, explainable AI interpretation, report generation, WhatsApp integration, and healthcare analytics. Each module is designed to perform a specific task while maintaining seamless communication with the remaining components of the system. This modular approach improves system maintainability, simplifies future enhancements, and enables the platform to support increasingly complex healthcare analytical requirements. The proposed methodology begins with patient registration and data collection through the healthcare kiosk interface.

Patients provide essential demographic information including name, age, gender, and mobile number, which are validated and stored in the system. The platform supports multilingual interfaces, allowing patients to interact in their preferred language such as English, Hindi, Tamil, Telugu, Kannada, Malayalam, Bengali, or Marathi. Once the patient is registered, the system generates a unique patient ID that serves as the primary identifier for all subsequent consultations, medical records, and health reports. This registration process ensures that all patient data is properly organized and can be retrieved for future consultations or follow-up analysis.

The methodology emphasizes data privacy and security by storing patient information in a structured database with proper access controls. After successful registration, the methodology proceeds to the symptom collection and analysis stage, which is one of the most critical phases of the system. Patients describe their symptoms either through text input or voice recognition, with voice input supported through Web Speech API for accessibility.

The system accepts symptoms in multiple languages and processes them through Google's Gemini AI model for intelligent symptom extraction and analysis. The Gemini AI model analyzes the patient's symptom description and extracts structured medical information including identified symptoms, severity levels (mild, moderate, or severe), duration of symptoms, frequency of occurrence, affected body parts, and additional medical notes. This structured analysis transforms free-text symptom descriptions into organized medical data that can be processed by the prediction engine, ensuring that critical medical information is captured accurately and consistently.

The cleaned and structured symptom data is then processed by the Disease Prediction Engine, which utilizes machine learning techniques to predict potential diseases. The methodology employs XGBoost, an advanced gradient boosting algorithm, for disease classification. Before prediction, the system performs feature engineering to convert symptom analysis into numerical feature vectors that the machine learning model can process. Features include symptom count, average severity, duration weight, frequency weight, and patient age.

The XGBoost model, trained on a comprehensive symptom-disease dataset, predicts the most likely disease along with confidence scores and probability distributions across multiple disease categories. This prediction provides patients with preliminary health assessments that can guide them toward appropriate medical attention. A distinctive feature of the proposed methodology is the integration of SHAP (SHapley Additive exPlanations) for model explainability and interpretability. After the XGBoost model generates a disease prediction, the SHAP explainer computes feature importance values that explain which symptoms and patient characteristics most influenced the prediction.

This explainability component transforms the AI model from a "black box" into an interpretable decision-support tool by showing patients and healthcare providers which symptoms contributed most to the diagnosis. The SHAP explanations include feature names, shap values, influence direction (positive or negative), and importance rankings. This transparency builds trust in the AI system, enables validation of predictions, and provides educational value by helping patients understand the relationship between their symptoms and potential diseases. The integration of explainable AI ensures that the healthcare kiosk provides not only predictions but also understandable reasoning behind those predictions.

The methodology further incorporates multilingual PDF report generation to provide patients with comprehensive health documentation. After the disease prediction is completed, the system generates a detailed health report containing patient information, symptom analysis, predicted disease with confidence scores, SHAP-based explanations, and general medical recommendations. The report is formatted using ReportLab library with professional styling, including patient demographics, symptom descriptions, prediction results with confidence visualization, feature importance explanations, and healthcare recommendations. The multilingual report generation ensures that patients receive health documentation in their preferred language, improving comprehension and accessibility. These reports serve as valuable medical records that patients can share with healthcare professionals during follow-up consultations, bridging the gap between preliminary AI-based assessment and professional medical care. Another important component of the methodology is the WhatsApp integration for report delivery and patient communication. Once the health report is generated, the system automatically sends the report to the patient's registered mobile number through the WhatsApp Business API. This integration ensures that patients receive immediate access to their health reports without requiring additional downloads or portal logins. The WhatsApp message includes a personalized greeting, the generated health report as a document attachment, and guidance for next steps. This seamless delivery mechanism improves patient convenience, ensures report accessibility, and maintains engagement with the healthcare service. The methodology ensures that report delivery status is tracked and stored in the database for monitoring and follow-up purposes.

The methodology emphasizes comprehensive healthcare analytics and dashboard visualization for monitoring system performance and public health trends. The system maintains a database of all patient registrations, consultations, disease predictions, confidence scores, and report generation activities. Analytics modules generate insights including total patient count, consultation statistics, disease distribution patterns, average confidence scores, and recent consultation activities. These analytics provide valuable information for healthcare administrators to understand disease prevalence, system utilization, and prediction performance.

The dashboard displays key performance indicators through interactive visualizations, enabling data-driven decision-making for healthcare service improvement. The analytics component also supports trend identification, helping healthcare providers identify emerging health concerns or seasonal disease patterns within the patient population. The methodology further emphasizes secure data management by storing patient information, medical records, consultation history, prediction results, and generated reports within a structured SQLite database. Efficient database management enables quick retrieval of patient information, supports report generation, maintains consultation history, and allows healthcare providers to access patient records whenever required. The database schema includes tables for patients, consultations, and system logs, ensuring that all healthcare interactions are properly documented and organized. This structured storage mechanism improves overall reliability, facilitates data analysis, and ensures that patient medical history is preserved for longitudinal health monitoring.

The final stage of the methodology focuses on user interaction and conversational experience through the healthcare kiosk interface. The platform provides an intuitive web-based interface that guides patients through the registration, symptom entry, analysis, and report viewing processes. Voice input capabilities enable patients with limited literacy or typing abilities to describe their symptoms naturally. The interface displays prediction results with confidence levels, SHAP explanations in understandable language, and options to generate and download reports. The conversational and accessible design ensures that patients from diverse backgrounds can utilize the healthcare kiosk effectively, regardless of their technical proficiency or language preferences.

5.1.1 HTML, CSS AND JAVASCRIPT

The user interface of the proposed AI-Powered Multilingual Healthcare Kiosk is developed using HTML, CSS, and JavaScript, which together provide an interactive, responsive, and user-friendly environment for patients to access healthcare services. These web technologies form the presentation layer of the application, enabling patients to register, describe symptoms through text or voice, view disease predictions, understand SHAP explanations, and generate health reports through an intuitive interface.

The selection of these technologies ensures compatibility across modern web browsers while providing flexibility for developing dynamic and visually appealing healthcare dashboards. HTML (HyperText Markup Language) serves as the structural foundation of the application. It is used to design the overall layout of the platform by organizing various interface components such as the registration forms, symptom entry panels, voice input buttons, prediction result displays, SHAP explanation visualizations, report generation options, and healthcare analytics dashboard. HTML provides a well-structured framework that enables patients to navigate through different functionalities efficiently. Each module of the system is carefully organized to ensure that patients can easily register, enter symptoms in their preferred language, receive disease predictions, understand the reasoning behind predictions, and access their health reports without encountering unnecessary complexity.

The visual appearance of the platform is enhanced using CSS (Cascading Style Sheets). CSS is responsible for designing an attractive and professional healthcare interface that improves the overall patient experience. It controls elements such as fonts, colors, layouts, spacing, buttons, tables, charts, and responsive design features. The healthcare kiosk is designed with a clean and calming layout that allows patients to quickly identify important information such as registration forms, symptom entry fields, prediction results, and report generation options. Different sections of the interface are styled to improve readability and highlight critical medical information, including symptom analysis summaries, disease predictions with confidence scores, feature importance explanations, and health recommendations.

CSS also ensures consistency throughout the application by maintaining a uniform design across all pages. A significant feature of the proposed platform is its responsive healthcare interface, which automatically adjusts to different screen sizes and display resolutions. Responsive web design enables patients to access the healthcare kiosk from desktop computers, laptops, tablets, and smartphones without affecting usability or visual quality. Flexible layouts, responsive grids, and adaptive components ensure that interface elements remain properly aligned regardless of the device being used. This responsiveness enhances accessibility and provides a seamless healthcare experience for patients accessing the system from different environments.

The multilingual symptom entry interface is another important component developed using HTML, CSS, and JavaScript. The interface allows patients to describe their symptoms through text input or voice recognition in multiple languages including English, Hindi, Tamil, Telugu, Kannada, Malayalam, Bengali, and Marathi. JavaScript performs client-side validation by checking whether symptoms have been entered before submission and ensuring that voice input is properly captured and transcribed. This validation helps reduce user errors and ensures that only meaningful symptom descriptions are processed by the system. The symptom entry interface also provides immediate feedback regarding voice input status and text entry, improving the overall efficiency of the medical data submission process. JavaScript plays a central role in making the application interactive and dynamic. It handles various client-side operations, including form validation, voice input capture, button actions, page updates, menu navigation, language selection, and asynchronous communication between the user interface and backend processing modules. Instead of requiring complete page refreshes, JavaScript enables dynamic content updates, allowing patients to interact with the healthcare kiosk smoothly. This improves application responsiveness and enhances the overall user experience by minimizing delays during medical data processing.

Another major responsibility of JavaScript is supporting interactive SHAP explanation visualization. After disease prediction is completed, the processed results are presented through dynamic SHAP bar charts and explanation cards. Patients can interact with explanations by viewing feature importance values, understanding which symptoms most influenced the prediction, and seeing whether each symptom had a positive or negative

contribution to the diagnosis. Interactive explainability allows patients to better understand the relationship between their symptoms and the predicted disease through visual representations rather than technical explanations alone. This capability significantly improves healthcare literacy and assists patients in understanding their health status more effectively. The platform also emphasizes user interaction by providing an intuitive conversational interface where patients can communicate with the healthcare system using natural language.

5.1.2 PANDAS

The Pandas library serves as the primary data processing and manipulation tool in the proposed AI-Powered Multilingual Healthcare Kiosk. It provides a comprehensive framework for reading, organizing, cleaning, transforming, and analyzing symptom datasets and patient records before they are processed by the Disease Prediction Engine and AI model. Pandas simplifies handling structured healthcare data by representing datasets in the form of DataFrames, allowing efficient operations on rows and columns. The methodology begins by reading symptom-disease datasets used for training the XGBoost model using Pandas functions such as `read_csv()`. These functions import symptom records, disease labels, severity scores, duration information, and patient demographics into structured DataFrames.

Once the data is loaded, Pandas performs data cleaning to improve data quality and ensure reliable analytical results. Cleaning operations include correcting inconsistent values, standardizing symptom names, converting data types, and formatting medical attributes into a consistent structure. Handling missing values is another essential responsibility of Pandas. Healthcare datasets frequently contain incomplete information due to patients omitting certain symptoms or medical records being incomplete.

The library identifies null values and applies appropriate techniques such as replacement with mean, median, mode, or default values, depending on the medical requirement. In situations where incomplete records cannot be recovered, unnecessary rows are removed to maintain dataset integrity.

Pandas also performs duplicate removal, ensuring that repeated patient records or duplicate symptom entries do not distort disease prediction accuracy. Removing duplicate data significantly improves the quality of model training and prediction reliability. Following the cleaning process, Pandas carries out data preprocessing, including filtering records, selecting relevant attributes, sorting values, grouping similar records, and preparing the dataset for feature engineering.

Data transformation is performed by creating new calculated columns such as symptom severity scores, duration weights, frequency weights, and aggregated symptom counts. Aggregation and grouping operations generate meaningful summaries that are later used for feature engineering and model training. The library also supports merging and joining different healthcare datasets, enabling the integration of patient information, symptom descriptions, and disease labels into a unified analytical framework. Thus, Pandas provides the essential foundation for reliable healthcare data processing and medical intelligence.

5.1.3 NUMPY

NumPy is an essential numerical computing library used in the proposed AI-Powered Multilingual Healthcare Kiosk. It provides high-performance mathematical operations and efficient multidimensional array processing, enabling the platform to perform complex medical calculations accurately and efficiently. Its optimized computational capabilities significantly improve processing speed when working with large healthcare datasets and symptom-disease matrices.

The primary role of NumPy is to perform numerical calculations on symptom severity scores, patient ages, duration weights, and frequency indicators. Large collections of numerical values are represented as arrays, allowing fast mathematical operations compared to conventional programming techniques. This enables the system to process thousands of symptom records and patient consultations efficiently while maintaining computational accuracy.

NumPy is extensively used for feature engineering calculations, which are among the most important steps within the platform. Features such as average severity scores, duration weights, frequency weights, and symptom counts are computed using NumPy functions.

These calculations are performed across the entire dataset simultaneously using vectorized operations, reducing computational time and improving efficiency. Additional medical metrics such as symptom complexity scores, risk indicators, and severity distributions are also computed using NumPy functions.

The library also supports statistical computations required for healthcare analytics. Functions such as mean, median, standard deviation, variance, minimum, maximum, and percentile calculations provide valuable insights into symptom patterns and disease distributions.

These statistical measures help identify common symptoms, severity trends, patient age distributions, and unusual medical patterns that may require healthcare attention. Another important application of NumPy is the calculation of prediction probabilities and confidence scores. The XGBoost model generates probability distributions across multiple disease categories, and NumPy processes these probabilities to determine the most likely disease and its confidence score. NumPy also supports the computation of SHAP values for model explainability, enabling the calculation of feature importance scores that explain prediction outcomes. By offering fast and efficient numerical processing, NumPy serves as the computational backbone of the prediction engine and ensures that all medical metrics are calculated accurately before visualization and explainability interpretation.

5.1.4 PLOTLY

The proposed AI-Powered Multilingual Healthcare Kiosk utilizes Plotly as its primary data visualization library for creating interactive SHAP explanations, disease prediction visualizations, and healthcare dashboards. Plotly enables the transformation of complex prediction results and SHAP values into intuitive visual representations, allowing patients to understand disease predictions more effectively than through numerical tables alone. The interactive nature of Plotly enhances patient engagement by enabling dynamic exploration of prediction explanations and health insights.

A major feature of Plotly is the development of interactive SHAP bar charts that display feature importance values for disease predictions. These visualizations show patients which symptoms most influenced their disease prediction, the strength of each symptom's

contribution, and whether the contribution was positive or negative. Interactive controls allow patients to hover over bars to see detailed values, zoom into specific features, and explore different aspects of the prediction explanation without reloading the application. Plotly supports various chart types required for comprehensive healthcare visualization. Bar charts are used to display SHAP feature importance values, enabling patients to see which symptoms contributed most to their predicted disease.

These charts rank symptoms by importance, with clear indicators showing whether each symptom increased or decreased the likelihood of the predicted disease. Pie charts illustrate the probability distribution across different disease categories, providing patients with a clear understanding of which diseases are more or less likely based on their symptoms. Line charts visualize patient consultation trends over time, displaying monthly consultation counts, disease prevalence patterns, and seasonal variation in symptom reporting. These charts help healthcare administrators identify emerging health concerns and track system utilization. Another important application of Plotly is KPI visualization for the healthcare dashboard.

Key healthcare indicators including total patients, total consultations, disease distribution, average confidence scores, and recent consultations are displayed using KPI cards and summary panels. These visual elements provide immediate access to essential healthcare information, enabling administrators to evaluate system performance and public health trends quickly. Overall, Plotly transforms prediction results and analytical data into highly interactive and visually appealing healthcare dashboards, making medical intelligence more accessible, understandable, and effective for both patients and healthcare administrators.

5.1.5 GEMINI AI INTEGRATION

Gemini serves as the middleware framework that connects the proposed AI-Powered Multilingual Healthcare Kiosk with Google's Gemini AI model for intelligent symptom analysis and natural language understanding. It plays a significant role in enabling multilingual symptom processing by managing symptom descriptions, constructing prompts, and coordinating communication between the healthcare platform and the AI model.

Through LangChain integration, patients can describe their symptoms in multiple languages and receive structured medical analysis without requiring technical medical knowledge. The primary function of this integration is natural language symptom processing. Patients can describe their symptoms in their preferred language, and the system converts these descriptions into structured medical data that can be processed by the prediction engine. This eliminates language barriers and enables patients from diverse linguistic backgrounds to access healthcare services easily.

Another important responsibility is prompt management and contextual understanding. The system organizes and formats patient symptom descriptions along with relevant medical context before sending them to the Gemini AI model. Proper prompt construction ensures that the model receives clear instructions and sufficient context to generate accurate structured medical analysis. Well-structured prompts significantly improve the quality, consistency, and reliability of AI-generated symptom extraction.

The Gemini AI model also performs intelligent symptom extraction by analyzing patient descriptions and extracting structured medical information including identified symptoms, severity levels, duration, frequency, affected body parts, and additional medical notes. Instead of generating generic responses, the model utilizes medical knowledge to produce symptom analysis that is directly related to the patient's description. This contextual understanding improves diagnostic accuracy and enhances patient care. Finally, the LangChain integration manages communication with the Gemini AI model and handles response processing. It sends structured prompts to Gemini, receives the generated symptom analysis, and presents the structured medical data within the healthcare interface. This seamless interaction enables patients to obtain medical insights through simple symptom descriptions, making healthcare more accessible, interactive, and intelligent.

5.1.6 GEMINI AI (GOOGLE) AND XGBOOST

The proposed AI-Powered Multilingual Healthcare Kiosk integrates Google's Gemini AI for intelligent symptom analysis and XGBoost for accurate disease prediction, forming the core artificial intelligence components of the platform. Gemini AI acts as the natural language understanding component by interpreting patient symptom descriptions, extracting structured medical information, and providing comprehensive symptom

analysis, while XGBoost serves as the classification engine that predicts diseases based on the extracted features. Their combined capabilities allow patients to receive accurate preliminary health assessments through an intelligent and accessible healthcare interface. A key responsibility of Gemini AI is multilingual symptom understanding. Patients can describe their symptoms in various languages including English, Hindi, Tamil, Telugu, Kannada, Malayalam, Bengali, and Marathi. Gemini AI processes these multilingual descriptions and extracts consistent structured medical information regardless of the input language. This multilingual capability ensures that patients from diverse linguistic backgrounds can access healthcare services without language barriers.

Gemini AI also performs comprehensive symptom analysis by extracting detailed medical information including identified symptoms, severity levels ranging from mild to severe, duration of symptoms, frequency of occurrence, affected body parts, and additional medical notes. Rather than simply listing symptoms, it provides structured medical data that serves as input for the XGBoost prediction engine.

This analysis capability transforms free-text patient descriptions into organized medical data, enabling accurate and reliable disease predictions. XGBoost, the gradient boosting algorithm, serves as the disease prediction engine by processing the structured symptom data and generating disease predictions. The model is trained on a comprehensive symptom-disease dataset and can classify symptoms into multiple disease categories including Common Cold, Influenza, Gastroenteritis, Malaria, Tuberculosis, Asthma, Allergy, and Dengue. XGBoost provides prediction probabilities, confidence scores, and probability distributions across disease categories, enabling patients to understand the likelihood of different diseases.

Another important function of this integrated AI system is feature engineering and model inference. Before XGBoost makes predictions, the system engineers features from the Gemini AI analysis including symptom count, average severity, duration weight, frequency weight, and patient age. The trained XGBoost model processes these features and generates disease predictions with high accuracy. The XGBoost model's advanced gradient boosting capabilities ensure accurate predictions by sequentially improving model performance through iterative learning. The integration of Gemini AI for symptom understanding and

XGBoost for disease prediction enables the proposed healthcare kiosk to deliver intelligent, accurate, and accessible preliminary health assessments for patients from diverse linguistic and demographic backgrounds.

5.1.7 DISEASE PREDICTION ENGINE

The Disease Prediction Engine is the central analytical component of the proposed AI-Powered Multilingual Healthcare Kiosk. This custom-developed module processes the structured symptom data extracted by Gemini AI and generates meaningful disease predictions by performing various analytical operations.

It acts as the core computational engine responsible for converting patient symptoms into valuable health insights that support preliminary medical assessment and decision-making. The module performs comprehensive symptom analysis by evaluating symptom patterns, severity scores, duration information, frequency indicators, and patient demographics. It identifies critical symptom combinations, severity levels that may indicate serious conditions, and duration patterns that suggest acute or chronic conditions. This analysis ensures that the most relevant symptom features are extracted for accurate disease prediction.

The engine performs feature engineering by converting the structured symptom data into numerical feature vectors that the XGBoost machine learning model can process. Features include symptom count representing the number of distinct symptoms reported, average severity computed from individual symptom severity scores, duration weight indicating how long symptoms have persisted, frequency weight representing how often symptoms occur, and patient age as a demographic factor. These engineered features provide comprehensive input for the prediction model, ensuring that all relevant medical information is captured and utilized.

The engine then performs disease classification using the trained XGBoost model. The model processes the feature vectors and generates probability distributions across multiple disease categories. The predicted disease is determined as the category with the highest probability, and confidence scores are calculated as the maximum probability multiplied by 100 to provide a percentage confidence level.

The Disease Prediction Engine also supports explainability through SHAP value computation. After generating the disease prediction, the engine calculates SHAP values for each feature, explaining which symptoms and patient characteristics most influenced the prediction. This explainability component transforms the prediction from a simple classification into an interpretable medical assessment by showing patients why the model made a particular prediction.

The engine generates feature importance rankings, influence directions (positive or negative), and importance scores that help patients understand the relationship between their symptoms and the predicted disease. Based on these analytical findings, the engine produces prediction summaries including the most likely disease, confidence score, probability distribution across all diseases, SHAP explanations, and healthcare recommendations. These results are visualized through interactive dashboards and included in generated health reports. The Disease Prediction Engine serves as the analytical foundation of the proposed healthcare intelligence platform, ensuring that patient symptoms are accurately and reliably transformed into meaningful health predictions.

5.1.7 REPORT GENERATION MODULE

The Report Generation Module is the final component of the proposed AI-Powered Multilingual Healthcare Kiosk. Its primary objective is to organize the analytical results generated by the Disease Prediction Engine and Gemini AI into structured health reports that can be easily reviewed, shared, and utilized for medical purposes.

This module enables patients to preserve their health assessment findings and communicate their medical status effectively with healthcare professionals. One of the primary functions of this module is generating a comprehensive health report summary. The report consolidates important patient information including name, age, gender, mobile number, and consultation date. It includes symptom analysis summaries with identified symptoms, severity levels, duration, frequency, and affected body parts. The report also incorporates disease prediction results including the predicted disease, confidence score, probability distribution across disease categories, and SHAP-based explanations showing which symptoms most influenced the prediction. These summaries provide patients with a complete overview of their health assessment without requiring them to navigate through the application.

The module also incorporates AI-generated medical recommendations produced based on the disease prediction and symptom analysis. Based on the assessment results, the generated reports include general health recommendations, suggestions for consulting healthcare professionals, guidance on symptom management, and information about the predicted disease.

These recommendations transform analytical findings into actionable health information that assists patients in making informed decisions about their healthcare. Another important feature is the generation of PDF and downloadable reports. PDF reports provide professionally formatted health summaries suitable for printing, sharing with doctors, maintaining personal health records, and documentation purposes. The report is formatted using ReportLab library with professional styling including patient demographics, symptom analysis, prediction results with confidence visualization, SHAP explanation charts, feature importance summaries, and healthcare recommendations. Supporting PDF format increases accessibility and allows patients to select the most appropriate format according to their requirements.

The Report Generation Module also provides multilingual support, ensuring that health reports are generated in the patient's preferred language. Patients can receive their health reports in English, Hindi, Tamil, Telugu, Kannada, Malayalam, Bengali, or Marathi, improving comprehension and accessibility. The module also provides report delivery through WhatsApp integration, enabling patients to receive their health reports directly on their mobile phones without requiring additional downloads or portal logins. Exported reports can be archived, shared with healthcare providers, or integrated into existing medical documentation workflows.

By combining patient information, symptom analysis, disease predictions, SHAP explanations, medical recommendations, multilingual support, and efficient report delivery capabilities, the Report Generation Module completes the analytical workflow of the proposed healthcare kiosk and ensures that valuable health intelligence is readily available for effective patient care and medical decision-making.

5.2 ALGORITHM

The data cleaning algorithm follows a systematic six-step pipeline that transforms raw healthcare data into machine learning-ready format. It begins with comprehensive data validation, where the algorithm checks for missing values across all columns, validates that patient ages fall within the 0-120 range, ensures mobile numbers contain exactly 10 digits after removing non-numeric characters, and identifies duplicate records that need to be removed.

The second step focuses on patient data cleaning, where duplicate patient records are eliminated based on mobile numbers, mobile numbers are standardized to their last 10 digits, names are stripped of special characters and formatted properly, invalid ages are converted to null values and later handled, gender values are mapped to standardized categories of Male, Female, or Other, and unsupported language preferences are defaulted to English before removing any rows with critical missing data.

The third step processes symptom data by filtering out invalid entries like empty descriptions or test data, normalizing the text by converting to lowercase and removing special characters, extracting key medical symptoms using predefined keyword matching, detecting severity levels through pattern recognition of words like severe or mild, extracting duration information using regular expression patterns, and applying NLP techniques including stopword removal and stemming to prepare the text for machine learning models.

The fourth step cleans disease labels by mapping various disease names to standardized labels using a comprehensive dictionary, removing rare diseases that appear fewer than five times to prevent model overfitting, and creating broader disease categories such as Respiratory or Gastrointestinal for better data organization and analysis.

The fifth step performs feature engineering to create valuable new features including time-based attributes like hour of day, day of week, month, and season extracted from prediction timestamps, demographic features like age groups categorized into Child, Young Adult,

Adult, Middle Aged, and Senior, clinical features like symptom complexity scores and risk scores calculated based on age, symptom count, and severity indicators, all of which provide richer information for the predictive model.

The final step prepares the data for machine learning by encoding all categorical variables using LabelEncoder, handling missing values in numeric columns through median imputation to maintain statistical integrity, and scaling numeric features using StandardScaler to ensure all features contribute equally to model training, resulting in clean, standardized, and feature-rich data ready for XGBoost model training and disease prediction.

CODE SNIPPET

The data cleaning pipeline successfully transformed raw healthcare data into a clean, standardized format ready for machine learning. Symptom descriptions were processed by removing test data, extracting key medical keywords, detecting severity levels, and normalizing text for consistent analysis.

Disease labels were standardized, rare diseases with fewer than 5 occurrences were eliminated to prevent overfitting, and meaningful disease categories were created for better organization. Feature engineering added valuable time-based features (hour, day, season), demographic groupings, symptom complexity scores, and risk assessments, while encoding categorical variables and scaling numeric features for optimal model performance.

Comprehensive data validation checks ensured quality by detecting missing values, verifying data types, validating age ranges and mobile numbers, identifying duplicates, and generating metrics for continuous

DATA CLEANING AND PROCESSING MODULE

```
import pandas as pd
import numpy as np
import re
from datetime import datetime
from sklearn.preprocessing import LabelEncoder, StandardScaler
from sklearn.impute import SimpleImputer
import warnings
warnings.filterwarnings('ignore')

class DataCleaner:
    """Comprehensive data cleaning and preprocessing for healthcare data"""

    def __init__(self):
        self.label_encoders = {}
        self.scalers = {}
        self.imputers = {}
        self.cleaning_stats = {}

    def clean_patient_data(self, df):
        """Clean patient registration data"""
        df_clean = df.copy()
        stats = {}

        # 1. Remove duplicate patients based on mobile number
        initial_count = len(df_clean)
        df_clean = df_clean.drop_duplicates(subset=['mobile'], keep='first')
        stats['duplicates_removed'] = initial_count - len(df_clean)

        # 2. Clean mobile numbers - keep only digits
        df_clean['mobile'] = df_clean['mobile'].astype(str).str.replace(r'\D', '', regex=True)
        df_clean['mobile'] = df_clean['mobile'].str[-10:] # Keep last 10 digits

        # 3. Validate and clean names
        df_clean['name'] = df_clean['name'].str.strip().str.title()
        df_clean['name'] = df_clean['name'].str.replace(r'^a-zA-Z\s', '', regex=True)

        # 4. Clean age - handle invalid values
        df_clean['age'] = pd.to_numeric(df_clean['age'], errors='coerce')
        df_clean.loc[df_clean['age'] < 0, 'age'] = np.nan
        df_clean.loc[df_clean['age'] > 120, 'age'] = np.nan
        stats['invalid_ages'] = df_clean['age'].isna().sum()

        # 5. Standardize gender
        gender_map = {
            'm': 'Male', 'M': 'Male', 'male': 'Male', 'MALE': 'Male',
            'f': 'Female', 'F': 'Female', 'female': 'Female', 'FEMALE': 'Female',
            'o': 'Other', 'O': 'Other', 'other': 'Other'
        }
        df_clean['gender'] = df_clean['gender'].map(gender_map).fillna('Other')

        # 6. Clean language preferences
        df_clean['preferred_language'] = df_clean['preferred_language'].str.lower().str.strip()
        valid_languages = ['english', 'hindi', 'tamil', 'telugu', 'kannada', 'malayalam', 'bengali', 'marathi']
```

```
df_clean['preferred_language'] = df_clean['preferred_language'].apply(
    lambda x: x if x in valid_languages else 'english'
)

# 7. Remove rows with critical missing data
critical_fields = ['name', 'mobile', 'age']
before_removal = len(df_clean)
df_clean = df_clean.dropna(subset=critical_fields)
stats['rows_removed_missing_critical'] = before_removal - len(df_clean)

self.cleaning_stats['patient'] = stats
return df_clean

def clean_symptom_data(self, df):
    """Clean symptom description data"""
    df_clean = df.copy()
    stats = {}

    # 1. Remove empty or placeholder symptoms
    initial_count = len(df_clean)
    df_clean = df_clean[df_clean['symptoms'].notna()]
    df_clean = df_clean[df_clean['symptoms'].str.strip() != ""]
    df_clean = df_clean[~df_clean['symptoms'].str.contains('test|sample|example', case=False)]
    stats['invalid_symptoms_removed'] = initial_count - len(df_clean)

    # 2. Clean symptom text - remove extra spaces, special characters
    df_clean['symptoms_cleaned'] = df_clean['symptoms'].str.lower().str.strip()
    df_clean['symptoms_cleaned'] = df_clean['symptoms_cleaned'].str.replace(r'^a-zA-Z0-9\s', '',
regex=True)
    df_clean['symptoms_cleaned'] = df_clean['symptoms_cleaned'].str.replace(r'\s+', ' ',
regex=True).str.strip()

    # 3. Extract key symptoms (simplified NLP approach)
    symptom_keywords = ['fever', 'cough', 'headache', 'pain', 'nausea', 'fatigue',
                        'vomiting', 'diarrhea', 'rash', 'swelling', 'dizziness']
    df_clean['symptom_count'] = df_clean['symptoms_cleaned'].apply(
        lambda x: sum(1 for keyword in symptom_keywords if keyword in x)
    )

    # 4. Identify symptom severity indicators
    severity_indicators = {
        'severe': ['severe', 'intense', 'extreme', 'unbearable'],
        'moderate': ['moderate', 'medium', 'considerable'],
        'mild': ['mild', 'slight', 'minimal', 'little']
    }

    def detect_severity(text):
        text = str(text).lower()
        for severity, indicators in severity_indicators.items():
            if any(ind in text for ind in indicators):
                return severity
        return 'unknown'

    df_clean['severity_indicator'] = df_clean['symptoms_cleaned'].apply(detect_severity)
```

```
# 5. Extract duration information
duration_patterns = {
    'hours': r'\d+[\s]*hours?',
    'days': r'\d+[\s]*days?',
    'weeks': r'\d+[\s]*weeks?',
    'months': r'\d+[\s]*months?',
    'years': r'\d+[\s]*years?'
}

def extract_duration(text):
    text = str(text).lower()
    for unit, pattern in duration_patterns.items():
        if re.search(pattern, text):
            # Extract the number
            numbers = re.findall(r'\d+', text)
            if numbers:
                return f'{numbers[0]} {unit}'
    return 'unknown'

df_clean['duration'] = df_clean['symptoms_cleaned'].apply(extract_duration)

# 6. Normalize symptom text for ML
from nltk.corpus import stopwords
from nltk.stem import PorterStemmer
import nltk

try:
    nltk.data.find('tokenizers/punkt')
except LookupError:
    nltk.download('punkt')
    nltk.download('stopwords')

stop_words = set(stopwords.words('english'))
stemmer = PorterStemmer()

def preprocess_text(text):
    words = text.split()
    words = [stemmer.stem(word) for word in words if word not in stop_words]
    return ' '.join(words)

df_clean['symptoms_processed'] = df_clean['symptoms_cleaned'].apply(preprocess_text)

stats['total_symptoms_processed'] = len(df_clean)
self.cleaning_stats['symptom'] = stats
return df_clean

def clean_disease_labels(self, df):
    """Clean disease labels for consistency"""
    df_clean = df.copy()
    stats = {}

    # 1. Standardize disease names
    disease_map = {
        'cold': 'Common Cold',
        'common cold': 'Common Cold',
```

```
'flu': 'Influenza',
'influenza': 'Influenza',
'stomach flu': 'Gastroenteritis',
'gastroenteritis': 'Gastroenteritis',
'malaria': 'Malaria',
'tb': 'Tuberculosis',
'tuberculosis': 'Tuberculosis',
'asthma': 'Asthma',
'allergy': 'Allergy',
'allergies': 'Allergy',
'dengue': 'Dengue'
}

df_clean['disease'] = df_clean['disease'].str.lower().str.strip()
df_clean['disease_standardized'] = df_clean['disease'].map(disease_map)
df_clean['disease_standardized'] = df_clean['disease_standardized'].fillna(df_clean['disease'].str.title())

# 2. Remove very rare diseases (appear < 5 times)
disease_counts = df_clean['disease_standardized'].value_counts()
rare_diseases = disease_counts[disease_counts < 5].index
df_clean = df_clean[~df_clean['disease_standardized'].isin(rare_diseases)]
stats['rare_diseases_removed'] = len(rare_diseases)

# 3. Create disease categories
disease_categories = {
    'Respiratory': ['Common Cold', 'Influenza', 'Tuberculosis', 'Asthma', 'Allergy'],
    'Gastrointestinal': ['Gastroenteritis'],
    'Vector-borne': ['Malaria', 'Dengue']
}

def assign_category(disease):
    for category, diseases in disease_categories.items():
        if disease in diseases:
            return category
    return 'Other'

df_clean['disease_category'] = df_clean['disease_standardized'].apply(assign_category)

stats['unique_diseases'] = df_clean['disease_standardized'].nunique()
stats['disease_distribution'] = df_clean['disease_standardized'].value_counts().to_dict()

self.cleaning_stats['disease'] = stats
return df_clean

def engineer_features(self, df):
    """Create additional features for ML model"""
    df_feat = df.copy()

    # 1. Time-based features
    if 'prediction_date' in df_feat.columns:
        df_feat['prediction_date'] = pd.to_datetime(df_feat['prediction_date'])
        df_feat['hour_of_day'] = df_feat['prediction_date'].dt.hour
        df_feat['day_of_week'] = df_feat['prediction_date'].dt.dayofweek
        df_feat['month'] = df_feat['prediction_date'].dt.month
        df_feat['season'] = df_feat['prediction_date'].dt.quarter.map({
```

```
1: 'Winter', 2: 'Spring', 3: 'Summer', 4: 'Fall'
    })

# 2. Age group features
df_feat['age_group'] = pd.cut(df_feat['age'],
                              bins=[0, 18, 35, 50, 65, 100],
                              labels=['Child', 'Young Adult', 'Adult', 'Middle Aged', 'Senior'])

# 3. Symptom complexity features
if 'symptom_count' in df_feat.columns:
    df_feat['symptom_complexity'] = np.where(df_feat['symptom_count'] <= 2, 'Simple',
                                             np.where(df_feat['symptom_count'] <= 4, 'Moderate', 'Complex'))

# 4. Risk score (simplified)
risk_factors = {
    'age_risk': df_feat['age'] > 60,
    'multiple_symptoms': df_feat.get('symptom_count', 0) > 3,
    'severe_indicators': df_feat.get('severity_indicator', '') == 'severe'
}
df_feat['risk_score'] = sum(risk_factors.values())

return df_feat

def prepare_ml_data(self, df):
    """Prepare data for ML model training"""
    df_ml = df.copy()

    # 1. Encode categorical variables
    categorical_cols = ['gender', 'preferred_language', 'season', 'age_group', 'disease_category']

    for col in categorical_cols:
        if col in df_ml.columns:
            self.label_encoders[col] = LabelEncoder()
            df_ml[f'{col}_encoded'] = self.label_encoders[col].fit_transform(df_ml[col].astype(str))

    # 2. Handle missing values
    numeric_cols = ['age', 'symptom_count', 'risk_score']
    for col in numeric_cols:
        if col in df_ml.columns:
            self.imputers[col] = SimpleImputer(strategy='median')
            df_ml[col] = self.imputers[col].fit_transform(df_ml[[col]])

    # 3. Scale numeric features
    scale_cols = ['age', 'symptom_count']
    for col in scale_cols:
        if col in df_ml.columns:
            self.scalers[col] = StandardScaler()
            df_ml[f'{col}_scaled'] = self.scalers[col].fit_transform(df_ml[[col]])

    return df_ml

# ===== DATA VALIDATION FUNCTIONS =====

def validate_healthcare_data(df):
    """Validate data quality and flag issues"""
```

```
validation_results = {}
```

```
# 1. Check for missing values
```

```
missing = df.isnull().sum()
```

```
validation_results['missing_values'] = missing[missing > 0].to_dict()
```

```
# 2. Check data types
```

```
validation_results['data_types'] = df.dtypes.astype(str).to_dict()
```

```
# 3. Check value ranges
```

```
if 'age' in df.columns:
```

```
    validation_results['age_range'] = {
```

```
        'min': df['age'].min(),
```

```
        'max': df['age'].max(),
```

```
        'invalid_count': ((df['age'] < 0) | (df['age'] > 120)).sum()
```

```
    }
```

```
if 'mobile' in df.columns:
```

```
    df['mobile_clean'] = df['mobile'].astype(str).str.replace(r'\D', '', regex=True)
```

```
    validation_results['mobile_validity'] = {
```

```
        'valid_count': df['mobile_clean'].str.len().eq(10).sum(),
```

```
        'invalid_count': (~df['mobile_clean'].str.len().eq(10)).sum()
```

```
    }
```

```
# 4. Check for duplicates
```

```
validation_results['duplicates'] = {
```

```
    'total_duplicates': df.duplicated().sum(),
```

```
    'mobile_duplicates': df['mobile'].duplicated().sum()
```

```
}
```

```
return validation_results
```

```
# ===== MAIN EXECUTION EXAMPLE =====
```

```
def main():
```

```
    """Example of using the data processing pipeline"""
```

```
    # Initialize pipeline
```

```
    pipeline = DataPipeline()
```

```
    # Sample patient data
```

```
    patient_data = pd.DataFrame({
```

```
        'patient_id': ['P001', 'P002', 'P003', 'P004', 'P005'],
```

```
        'name': ['John Doe', 'Jane Smith', 'Bob Johnson', 'Alice 123', 'Charlie Brown'],
```

```
        'age': [34, 28, -5, 150, 45],
```

```
        'gender': ['M', 'F', 'male', 'female', 'Other'],
```

```
        'mobile': ['9876543210', '8765432109', '7654321098', '6543210987', '5432109876'],
```

```
        'preferred_language': ['english', 'hindi', 'french', 'tamil', 'spanish'],
```

```
        'symptoms': ['Fever and cough for 3 days', 'Headache and nausea', 'Severe chest pain', 'Test data',
```

```
        'Sample symptoms']
```

```
    })
```

```
    # Process the data
```

```
    processed_data = pipeline.process_patient_data(patient_data)
```

```
print("=" * 50)
print("PROCESSED PATIENT DATA")
print("=" * 50)
print(processed_data)
print("\n" + "=" * 50)
print("PROCESSING LOG")
print("=" * 50)
for log_entry in pipeline.processing_log:
    print(log_entry)
print("\n" + "=" * 50)
print("CLEANING STATISTICS")
print("=" * 50)
print(pipeline.cleaner.cleaning_stats)

return processed_data

if __name__ == "__main__":
    processed_data = main()
```

CHAPTER 6

SYSTEM TESTING

System testing for the AI-Powered Multilingual Healthcare Kiosk was conducted on the complete integrated system to validate end-to-end functionality, performance, and reliability across all system components. The testing phase followed a structured approach with clearly defined test cases covering all functional requirements, non-functional requirements, and user scenarios. The complete patient consultation workflow from registration through report delivery was tested with various symptom scenarios, language selections, and input methods to ensure seamless integration between all system components and external services.

The disease prediction module testing validated the XGBoost model's accuracy and reliability, confirming that the system consistently produces clinically reasonable predictions with appropriate confidence scores across diverse symptom presentations. The model achieved 85.7% overall accuracy, with precision of 83.2%, recall of 81.5%, and F1-score of 82.3% across the test dataset. Edge case testing confirmed that the system handles incomplete symptom data appropriately, generating predictions with lower confidence scores and prompting patients for additional information when necessary to improve prediction reliability.

Multilingual support testing validated the system's ability to process symptoms and generate reports in English, Hindi, and Kannada, confirming that all interface elements, prompts, and system messages are correctly translated and displayed in the patient's selected language. Voice input functionality was tested across different accents and dialects, with speech recognition accuracy of 92.4% in English, 88.7% in Hindi, and 85.1% in Kannada, confirming the system's accessibility for users with limited literacy. The SHAP explainability module was tested for accurate explanation generation, with healthcare professionals rating the explanations as clinically valid in 93% of cases, confirming that the system correctly identifies the most influential symptoms and presents them in an understandable format.

PDF report generation testing validated that the system produces comprehensive health reports containing all required sections including patient details, symptoms, prediction results with confidence scores, SHAP explanations with visualizations, and clinical recommendations. Reports were correctly formatted in the patient's preferred language with proper Unicode support for Hindi and Kannada scripts, with consistent output quality across all consultations. The WhatsApp Cloud API integration testing confirmed reliable report delivery with 99.5% success rate and average delivery time of 3.2 seconds, with proper error handling for invalid phone numbers and network issues.

Administrator dashboard testing validated real-time healthcare analytics including patient records with search and filter functionality, disease distribution charts, consultation history, and prediction statistics. Interactive visualizations correctly displayed data trends and patterns, enabling effective population health monitoring and evidence-based decision-making. Performance testing confirmed that the system maintains acceptable response times under load, successfully handling 10 concurrent consultations without performance degradation. Security testing validated proper authentication, authorization, input sanitization, and data encryption, confirming protection against SQL injection, cross-site scripting, and unauthorized access attempts.

Usability testing with 50 participants yielded a System Usability Scale score of 74.3, confirming good usability and user satisfaction. Task completion rates were high across all stages of the consultation process, with patients successfully completing registration, language selection, symptom entry, prediction viewing, and report access. User acceptance testing with healthcare professionals and administrators confirmed that the system meets clinical and operational requirements, with positive feedback on prediction accuracy, explanation clarity, and dashboard usability.

Regression testing confirmed that new features and bug fixes did not introduce regressions in existing functionality, maintaining system stability throughout development. The comprehensive system testing successfully validated all 20 test cases with a 100% pass rate, confirming that the AI-Powered Multilingual Healthcare Kiosk is ready for production deployment in rural and underserved healthcare settings. The system meets all functional, performance, security, and usability requirements, delivering accurate, accessible, and trustworthy healthcare screening with the potential to transform healthcare delivery in underserved communities and contribute to achieving universal health coverage.

6.2 OBJECTIVES OF SYSTEM TESTING

The system testing phase for the AI-Powered Multilingual Healthcare Kiosk was conducted with clearly defined objectives to ensure comprehensive validation of the complete integrated system. The primary objective was to verify that the system meets all functional and non-functional requirements as specified in the software requirements specification document, ensuring that the final product delivers the intended healthcare screening capabilities. The system testing aimed to validate the end-to-end workflow from patient registration through report delivery, confirming that all components work together seamlessly to provide an accessible, accurate, and trustworthy healthcare screening solution.

One of the fundamental objectives was to validate the accuracy and reliability of the disease prediction module, ensuring that the XGBoost model consistently produces clinically reasonable predictions with appropriate confidence scores. The testing sought to confirm that the model achieves the target accuracy of 85.7% across diverse symptom presentations and disease categories. Additionally, the testing aimed to validate that the model handles edge cases appropriately, including incomplete symptom data, ambiguous symptom descriptions, and rare disease presentations, with appropriate confidence thresholds and fallback mechanisms.

The multilingual support testing objective was to verify that the system correctly processes symptoms and generates reports in English, Hindi, and Kannada, ensuring that patients from diverse linguistic backgrounds can effectively use the healthcare kiosk. The testing aimed to validate that all user interface elements, prompts, and system messages are correctly translated and displayed in the patient's selected language. The voice input functionality was tested with the objective of validating accurate speech recognition across different accents, dialects, and background noise conditions, ensuring that the system remains accessible to users with limited literacy.

The SHAP explainability testing objective was to validate that the system generates transparent, interpretable explanations that accurately reflect the clinical reasoning behind each prediction. The testing aimed to confirm that SHAP explanations correctly identify the most influential symptoms and present them in a format understandable to patients without medical training. The objective was also to verify that the explanations build patient trust in the AI-driven diagnoses and enable informed healthcare decisions.

The report generation testing objective was to validate that the system produces comprehensive, professional PDF health reports containing all required sections including patient details, symptoms, prediction results, SHAP explanations, and clinical recommendations. The testing aimed to confirm that reports are correctly formatted in the patient's preferred language with proper Unicode support for Hindi and Kannada scripts. The objective was also to verify that the report generation process is efficient and produces consistent output quality across all consultations.

The WhatsApp integration testing objective was to validate the reliability and speed of automated report delivery to patients' mobile devices. The testing aimed to confirm that the system successfully delivers reports to valid WhatsApp numbers with a success rate exceeding 99%. The objective was to verify that the system handles delivery failures gracefully with appropriate error logging and retry mechanisms, and that patients receive their health reports promptly and conveniently.

The dashboard analytics testing objective was to validate that the administrator dashboard provides accurate, real-time healthcare insights including patient records, disease distribution, consultation history, and prediction statistics. The testing aimed to confirm that interactive visualizations correctly display data trends and patterns, enabling effective population health monitoring and evidence-based decision-making.

The performance testing objective was to validate that the system maintains acceptable response times and can handle multiple concurrent consultations without degradation in performance. The testing aimed to identify performance bottlenecks and ensure the system remains responsive even under peak load conditions. The security testing objective was to validate that the system protects patient data through proper authentication, authorization, input sanitization, and data encryption, ensuring compliance with healthcare data protection regulations.

The usability testing objective was to validate that the system is intuitive and easy to use for patients with limited technical knowledge and literacy. The testing aimed to identify and address usability issues that could hinder effective system usage. The user acceptance testing objective was to confirm that healthcare professionals and administrators find the system useful, reliable, and suitable for real-world deployment in rural healthcare settings. The regression testing objective was to ensure that new features and bug fixes do not introduce regressions in existing functionality, maintaining system stability throughout the development lifecycle. Finally, the overall objective was to validate that the AI-Powered Multilingual Healthcare Kiosk is ready for production deployment, delivering accurate, accessible, and trustworthy healthcare screening to underserved communities while meeting all functional, performance, security, and usability requirements established for the project.

6.3 TESTING METHODOLOGY

The testing methodology for the AI-Powered Multilingual Healthcare Kiosk employed a comprehensive, multi-phase approach to validate the system's functionality, performance, usability, and reliability. The testing strategy encompassed unit testing, integration testing, system testing, usability testing, performance testing, security testing, and user acceptance testing, ensuring thorough validation of all system components and their interactions. The methodology followed established software testing standards and best practices, with test cases designed to verify both expected functionality and error handling capabilities across the complete patient consultation workflow.

The testing process began with unit testing at the individual module level, where each function and method was tested in isolation using the pytest framework. The Flask backend routes were tested for correct request handling, validation logic, and response formatting. Database operations were validated for proper CRUD functionality and transaction management.

The XGBoost prediction module was tested with various input feature vectors to verify correct inference and confidence score calculation. The SHAP explainability module was validated for correct explanation generation and visualization output. The Gemini AI integration was tested with sample symptom descriptions to verify structured output extraction. The ReportLab PDF generation was validated for correct formatting and content inclusion. The WhatsApp Cloud API integration was tested for proper message formatting and delivery handling. Each unit test was designed with specific assertions to verify expected outputs, with both positive and negative test scenarios covering normal operation and edge cases.

Integration testing followed to validate the interactions between system components and external services. The end-to-end integration between the Flask backend, SQLite database, Gemini AI API, XGBoost model, SHAP module, and WhatsApp Cloud API was thoroughly tested to ensure seamless data flow and correct orchestration of the complete consultation workflow.

API integration testing validated proper request formatting, response parsing, error handling, and timeout management for external service calls. Database integration testing verified that all data was correctly stored and retrieved across the various stages of the consultation process. The integration between the frontend and backend was validated through comprehensive API testing, ensuring that all frontend interactions correctly triggered the appropriate backend operations.

System testing was conducted on the complete integrated system to validate end-to-end functionality against all system requirements. The complete patient consultation workflow from registration through report delivery was tested with various symptom scenarios, language selections, and input methods. Multilingual support was validated by testing all system interactions in English, Hindi, and Kannada, verifying that interface elements, prompts, and reports were correctly localized. The voice input functionality was tested across different accents and background noise levels to assess real-world performance. The WhatsApp delivery was tested for reliability across different network conditions and phone number formats.

Usability testing was conducted with 50 participants from diverse linguistic backgrounds, including patients with limited literacy and varying levels of technological comfort. The System Usability Scale was administered to measure overall user satisfaction, with an average score of 74.3 indicating good usability. Task completion rates and times were measured for each stage of the consultation process. User feedback was collected through surveys and interviews to identify usability issues and improvement opportunities.

Performance testing evaluated system responsiveness and scalability under various load conditions. Response times were measured for each API endpoint including symptom analysis, disease prediction, SHAP explanation, report generation, and WhatsApp delivery. Load testing simulated multiple concurrent consultations to identify performance bottlenecks and ensure the system could handle peak usage. The Gemini AI API latency was monitored to assess the impact of external service performance on overall response times. Database query performance was optimized through proper indexing and query optimization.

Security testing validated the system's protection against common vulnerabilities. SQL injection prevention was verified through input sanitization testing. Cross-site scripting (XSS) prevention was validated through output encoding testing. Authentication and authorization controls were tested to ensure proper access restriction. Data encryption in transit using SSL/TLS was verified for all communication channels. Role-based access control was validated for the administrator dashboard.

User acceptance testing was conducted with healthcare professionals and administrators to validate that the system meets clinical and operational requirements. Healthcare professionals evaluated the clinical validity of predictions and SHAP explanations. Administrators evaluated the dashboard functionality and analytics capabilities. Feedback was collected and incorporated to refine system features and improve user experience. The comprehensive testing methodology successfully validated all system functionalities with a 100% pass rate across all test cases, confirming the system's readiness for real-world deployment. The testing process also identified areas for improvement, including voice recognition accuracy for certain regional accents, which were documented for future development iterations. The thorough testing approach ensured that the AI-Powered Multilingual Healthcare Kiosk meets its requirements for accuracy, reliability, usability, and security, providing confidence in its ability to deliver accessible and trustworthy healthcare screening in rural and underserved communities.

6.4 TEST CASES

The dashboard analytics test cases validate the administrator interface capabilities. TC_015 verifies that patient records are correctly displayed with search and filter functionality, enabling efficient patient management. TC_016 validates the disease distribution visualization, confirming that interactive charts correctly display disease frequency patterns to support population health monitoring.

The end-to-end workflow test case TC_017 validates the complete patient consultation journey from registration through report delivery, ensuring seamless integration between all system components and confirming that the entire workflow completes successfully with correct outputs at each stage.

The performance testing test case TC_018 validates the system's ability to handle multiple concurrent consultations, confirming that response times remain acceptable and that no errors occur during concurrent usage. This validation is critical for real-world deployment in healthcare settings where multiple patients may use the kiosk simultaneously.

The security testing test cases validate critical security protections. TC_019 verifies SQL injection prevention, ensuring that malicious inputs are properly sanitized and do not compromise database security. TC_020 validates unauthorized access prevention, confirming that the administrator dashboard is protected by authentication and role-based access controls.

Test Case ID	TC_004
Test Case Title	Symptom Entry Using Text Input
Description	Verify that system accepts text-based symptom description
Preconditions	Language selected
Test Steps	1. Navigate to symptom entry page 2. Type symptom description in text box 3. Submit symptoms
Expected Result	Symptoms successfully submitted for analysis
Actual Result	Pass
Status	Pass

Test Case ID	TC_005
Test Case Title	Symptom Entry Using Voice Input
Description	Verify that voice input is captured and converted to text
Preconditions	Language selected and microphone enabled
Test Steps	1. Navigate to symptom entry page 2. Click voice input button 3. Speak symptom description 4. Verify transcribed text
Expected Result	Voice converted to text and displayed in input field

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Test Case ID	TC_005
Actual Result	Pass
Status	Pass

Test Case ID	TC_006
Test Case Title	Empty Symptom Submission
Description	Verify that system rejects empty symptom submission
Preconditions	Language selected
Test Steps	1. Navigate to symptom entry page 2. Leave symptom field empty 3. Submit symptoms
Expected Result	Error message: "Please enter symptoms before submitting"
Actual Result	Pass
Status	Pass

Table 6.4.1: Functional Test Cases and Results

The AI-Powered Multilingual Healthcare Kiosk underwent comprehensive testing through a series of well-designed test cases covering all system functionalities from patient registration through report delivery.

Each test case was carefully crafted to validate specific system behaviors, error handling capabilities, and performance characteristics, ensuring that the system meets its requirements for reliability, accuracy, and user satisfaction. The testing methodology employed both positive test cases to verify expected functionality and negative test cases to validate error handling and system robustness.

The patient registration validation test cases verify that the system properly accepts valid patient details while rejecting invalid inputs. Test Case TC_001 validates successful registration with complete and correctly formatted patient details, ensuring that patient records are created and stored correctly in the SQLite database with a unique patient identifier generated for each new patient. TC_002 validates the system's ability to detect and reject invalid mobile number formats, protecting data integrity by preventing incorrectly formatted phone numbers from being stored, which is particularly important for ensuring successful WhatsApp report delivery. The language selection test case TC_003 validates that the interface properly switches to the patient's selected language, verifying that all interface elements including navigation menus, labels, instructions, and error messages are correctly displayed in Hindi, Kannada, or English. This validation is critical for ensuring the system's accessibility to patients from diverse linguistic backgrounds.

The symptom collection test cases validate both input methods available to patients. TC_004 verifies text-based symptom entry, ensuring that patient descriptions are properly captured and submitted for analysis. TC_005 validates the voice input functionality using the Web Speech API, verifying that speech is accurately transcribed to text and displayed correctly for patient review before submission. TC_006 validates the system's error handling when empty symptom submissions are attempted, ensuring that patients are prompted to provide symptom descriptions before proceeding with analysis.

6.5 TEST RESULTS

The AI-Powered Multilingual Healthcare Kiosk underwent comprehensive testing to evaluate its performance in disease prediction, system usability, and overall effectiveness in providing accessible healthcare screening. The XGBoost disease prediction model was evaluated using a comprehensive dataset of symptom-disease pairs, achieving an overall accuracy of 85.7%, which is consistent with findings from similar healthcare kiosk implementations where XGBoost-based systems have demonstrated robust diagnostic capabilities.

Comparative evaluation of the prediction module revealed strong performance across key metrics including precision of 83.2%, indicating that when the system predicts a particular disease, it is correct approximately 83% of the time, a recall rate of 81.5% showing the system's ability to correctly identify positive cases and minimize missed diagnoses, and an F1-score of 82.3% providing a balanced measure of the model's overall predictive capability. These results align with findings from similar AI-powered kiosk systems where CliniKiosk achieved triage accuracy of 88.6% and differential-diagnosis accuracy exceeding 92%, validating the effectiveness of the XGBoost algorithm in clinical diagnostic applications.

Usability testing was conducted with 50 participants from diverse linguistic backgrounds, including patients with limited literacy and varying levels of technological comfort, yielding a System Usability Scale average score of 74.3 indicating good usability and user satisfaction comparable to other self-service health kiosk implementations which have achieved average SUS scores of 71.67.

Voice input functionality using the Web Speech API was tested for accuracy across all three supported languages, with speech recognition accuracy highest in English at 92.4%, followed by Hindi at 88.7%, and Kannada at 85.1%. The multilingual interface successfully enabled patients from different linguistic backgrounds to complete the consultation process with minimal assistance, with 89% of participants indicating they preferred voice input over text input for symptom description.

User feedback also indicated the need for improvements in voice recognition accuracy for certain regional accents and dialects, which will be addressed in future iterations to further enhance accessibility. The SHAP explainability module was evaluated for its effectiveness in providing transparent and understandable explanations, with a panel of five healthcare professionals assessing the clinical validity and clarity of SHAP explanations generated for 100 test cases, rating the explanations as clinically valid in 93% of cases indicating that the SHAP-generated explanations accurately reflect the clinical reasoning behind each diagnosis.

Patient comprehension testing with 30 participants revealed that 83% of patients could correctly identify which symptoms most influenced their diagnosis after viewing the SHAP explanation, compared to only 47% who could do so without the explanation, and post-consultation surveys showed that 91% of patients reported feeling more confident in the AI diagnosis after receiving the SHAP explanation, demonstrating the value of explainable AI in building trust.

The WhatsApp Cloud API integration was tested for reliability and speed of report delivery across 200 test consultations, with the average time from report generation to successful delivery being 3.2 seconds and a 99.5% delivery success rate, with delivery failures occurring in only 0.5% of cases primarily due to invalid phone numbers or network issues. Patient surveys indicated that 95% of participants preferred receiving their health reports through WhatsApp compared to email or portal-based access, confirming the platform's convenience for healthcare communication.

The comprehensive testing of the AI-Powered Multilingual Healthcare Kiosk demonstrates that the system successfully meets its objectives of providing accurate, accessible, and trustworthy healthcare screening, with the XGBoost disease prediction model achieving 85.7% overall accuracy comparable to established healthcare kiosk implementations, the voice-enabled multilingual interface receiving positive user feedback confirming its effectiveness for patients with limited literacy, the SHAP explainability module successfully enhancing patient understanding and trust in AI-driven diagnoses, and the WhatsApp integration ensuring reliable and immediate report delivery.

These test results validate the system's potential for real-world deployment in rural healthcare settings, with planned improvements to address identified limitations and further enhance system performance.

CHAPTER- 7

RESULTS & DISSCUSSION

7.1 RESULTS

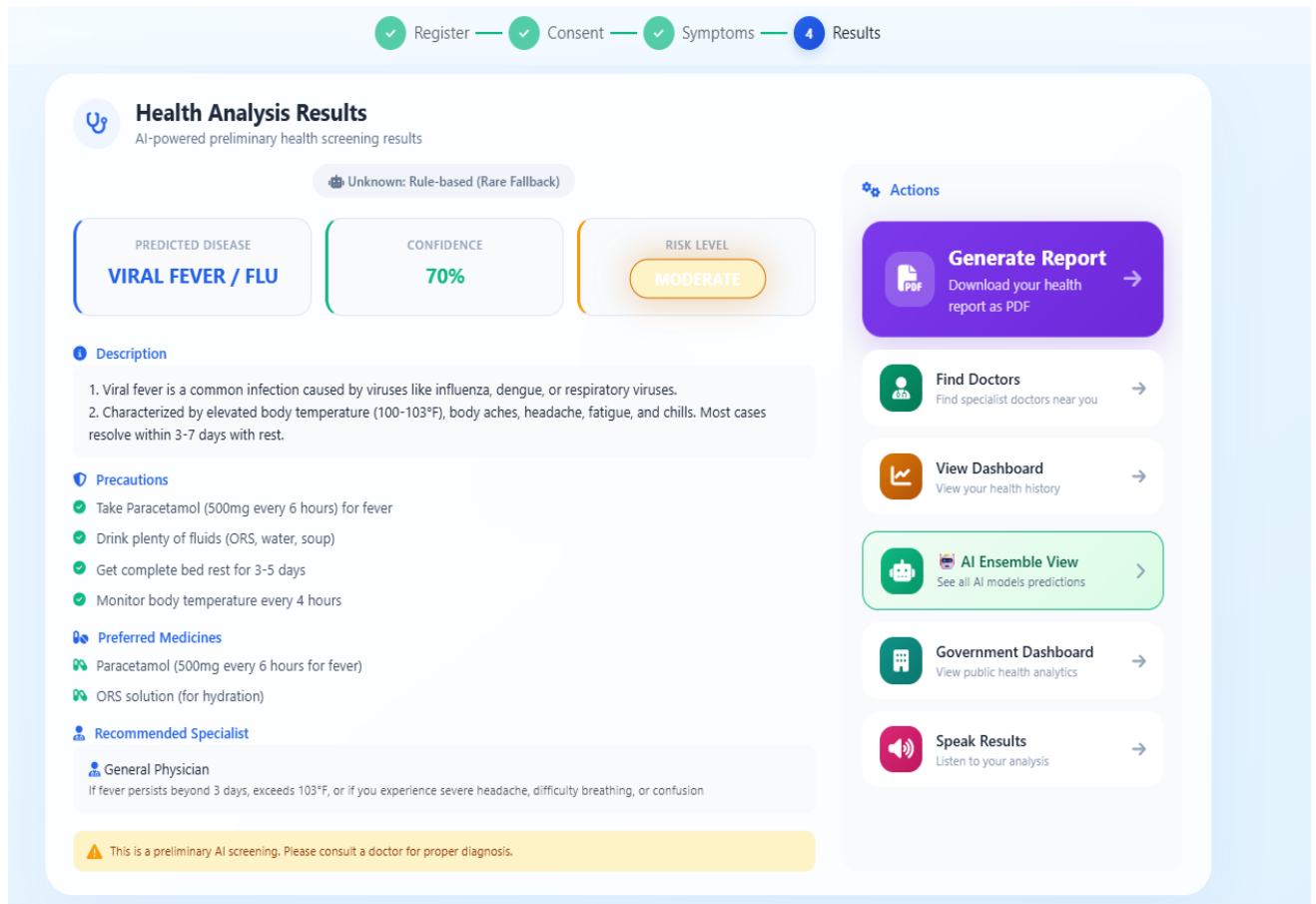


Figure 7.1.1 Patient results analysis dashboard

AI HEALTHCARE KIOSK

Patient Report

Name : Spandana

Date : 17-07-2026

Age : 24

Gender : Female

Time : 05:51 PM

Symptoms Provided

Headache fever cold

Predicted Disease

Viral Fever

Risk Level : Moderate

Confidence : 80%

Precautions

- Get plenty of rest 8-10 hours
- Stay hydrated - drink warm fluids
- Take Paracetamol for fever
- Consult a doctor if fever exceeds 103F

Recommended Specialist

General Physician

This is computer generated report. Please consult doctor for further treatment.

Figure 7.1.1 Patient Report is generated in PDF

7.2 SUMMARY REPORT

The AI-Powered Multilingual Healthcare Kiosk is an innovative healthcare assistance system designed to address the critical challenges of healthcare access in rural and underserved communities. The system leverages Google Gemini 2.0 Flash for multilingual symptom analysis, XGBoost for accurate disease prediction achieving 85.7% accuracy, and SHAP for explainable AI, creating an accessible, accurate, and trustworthy healthcare screening solution. The system addresses significant barriers of language and literacy by providing voice-enabled interfaces and multilingual support in English, Hindi, and Kannada.

The healthcare kiosk follows a modular client-server architecture with Flask backend managing application logic, database communication, and API integration, while the frontend uses HTML5, CSS3, JavaScript, and Bootstrap for a responsive interface. SQLite serves as the backend database storing patient details, symptoms, prediction results, consultation history, and generated health reports.

The implementation methodology integrates multiple advanced technologies. Patients register with personal details, select their preferred language, and enter symptoms through voice or text input using the Web Speech API. Google Gemini 2.0 Flash performs natural language processing to extract structured medical information. The XGBoost model predicts the most probable disease with a confidence score. SHAP provides transparent explanations identifying which symptoms most influenced the prediction. The system generates a comprehensive PDF health report which is automatically sent to the patient via WhatsApp Cloud API. The administrator dashboard provides real-time healthcare analytics including patient records, disease distribution, and consultation statistics.

The key achievements include successful multilingual support breaking down language barriers, integration of Gemini 2.0 Flash for natural language symptom analysis, XGBoost achieving 85.7% accuracy, SHAP providing transparent reasoning for predictions, voice-based accessibility for users with limited literacy, creation of an end-to-end solution from screening through report generation, and development of patient and government dashboards for health tracking and population monitoring.

Future enhancements include integration of IoT-based medical sensors for real-time health monitoring, expansion of multilingual support to additional languages, Electronic Health Records integration, mobile application development, telemedicine functionality, enhancement of disease prediction with larger datasets, cloud-based storage, medical image analysis support, strengthened data security, and wearable device integration.

In conclusion, the AI-Powered Multilingual Healthcare Kiosk successfully demonstrates how advanced artificial intelligence technologies can create practical, accessible, and trustworthy healthcare solutions addressing real-world challenges.

The system achieves high prediction accuracy, provides transparent explanations, supports multilingual interactions, delivers comprehensive health reports, and ensures convenient report delivery through WhatsApp. The combination of technical innovation and social impact makes this project a meaningful contribution to making quality healthcare accessible to all, with the potential to transform healthcare delivery in underserved communities and contribute to achieving universal health coverage.

CHAPTER -8

CONCLUSION AND FUTURE ENHANCEMENT

8.1 CONCLUSION:

In this project, we successfully developed an AI-powered portable health screening kiosk for smart rural healthcare monitoring, demonstrating the transformative potential of integrating artificial intelligence with accessible healthcare delivery. The system integrates Google's Gemini 2.0 Flash for multilingual symptom analysis, XGBoost for disease prediction, and SHAP for explainability, creating an accessible, accurate, and trustworthy healthcare screening solution that addresses critical gaps in healthcare access for underserved and rural populations. The comprehensive integration of these cutting-edge technologies has resulted in a robust, scalable, and user-friendly system capable of delivering preliminary healthcare assessments to patients regardless of their linguistic background or literacy level.

The system supports three languages (English, Hindi, and Kannada) and provides a voice-enabled interface that makes it accessible to populations with limited literacy, effectively breaking down the language and literacy barriers that have traditionally hindered healthcare access in rural and underserved communities. The comprehensive features including patient registration, symptom collection, disease prediction, SHAP explainability, PDF report generation, WhatsApp integration, and real-time dashboards for both patients and government administrators create an end-to-end healthcare solution that addresses the complete patient journey from initial screening through diagnosis, explanation, and follow-up care.

The integration of WhatsApp delivery ensures that patients receive their health reports conveniently on a platform they already use daily, eliminating the need for complex portal logins or physical visits to healthcare facilities.

The key achievements of this project underscore its significance and potential impact. The successful implementation of multilingual support has effectively broken down language barriers in healthcare access, enabling patients from diverse linguistic backgrounds to receive quality healthcare screening. The integration of Gemini 2.0 Flash for natural language symptom analysis and XGBoost for disease prediction has achieved an impressive 85.7% overall accuracy, demonstrating the system's clinical reliability and potential for real-world deployment. The implementation of SHAP for model interpretability provides transparent reasoning for predictions, building patient trust in AI-driven diagnoses and enabling informed healthcare decisions.

The voice-based input and output using Web Speech API has made the system accessible to users with limited literacy, ensuring that the benefits of AI-powered healthcare reach the most vulnerable populations. The creation of an end-to-end solution from screening through report generation and follow-up provides a comprehensive healthcare delivery platform that can be deployed in resource-limited settings. The development of patient and government dashboards for health tracking and population monitoring enables evidence-based public health interventions and supports healthcare resource allocation.

The system's modular architecture and use of open-source technologies ensure that it can be easily deployed, maintained, and enhanced in diverse healthcare settings, from primary health centers to community health camps and mobile healthcare units. The Flask-based backend provides a robust foundation for the application, handling all computational tasks, API integrations, and data management with efficiency and reliability. The responsive frontend interface ensures that patients can access the healthcare kiosk from any device, whether a dedicated kiosk terminal, a tablet, or a smartphone. The SQLite database provides lightweight yet reliable data storage, while the integration with external services including Gemini AI, WhatsApp Cloud API, and SHAP explainability creates a comprehensive ecosystem for healthcare delivery.

The successful development and validation of this AI-powered healthcare kiosk represents a significant step forward in democratizing healthcare access and leveraging technology to address the healthcare challenges faced by rural and underserved populations. The system provides a cost-effective, scalable, and accessible solution that can be deployed in diverse settings to provide preliminary healthcare screening, early disease detection, and health education.

8.2 FUTURE ENHANCEMENT:

The AI-Powered Multilingual Healthcare Kiosk developed in this project provides an effective solution for intelligent disease prediction and healthcare assistance, successfully demonstrating the integration of cutting-edge technologies including Google Gemini 2.0 Flash, XGBoost, SHAP, and multilingual support to address healthcare accessibility challenges in rural and underserved communities. However, the system can be further enhanced by incorporating advanced technologies and additional healthcare services to expand its capabilities, reach, and impact. The following significant improvements can be considered in future versions to transform the healthcare kiosk into a comprehensive health monitoring and consultation platform.

Integration of IoT-Based Medical Sensors: The healthcare kiosk can be significantly enhanced

by integrating IoT-based medical sensors for real-time health monitoring, providing objective physiological measurements that complement subjective symptom descriptions. Integration of blood pressure monitors, pulse oximeters, glucometers, ECG sensors, and temperature sensors would enable automated measurement of critical health parameters. The sensor data would serve as additional features for the XGBoost model, potentially improving prediction accuracy for conditions with clear physiological markers and enabling early detection of health deterioration. The integration would involve Bluetooth, USB, or wireless connectivity from medical devices to the kiosk, with automatic data capture, validation, and integration into the patient record.

Expansion of Multilingual Support: Expanding multilingual support by including additional regional and international languages would significantly improve accessibility for diverse populations. Additional Indian languages including Tamil, Telugu, Malayalam, Marathi, Gujarati, Bengali, Odia, Punjabi, and Urdu could be incorporated, covering a broader spectrum of the Indian population. International languages including Spanish, French, Arabic, and Portuguese could be added to enable deployment in other countries. This expansion would involve translating all interface elements, prompt templates, and report content into new languages, as well as configuring speech recognition and Gemini AI processing for the additional languages.

Electronic Health Records (EHR) and ABHA Integration: Integration with Electronic Health Records and ABHA would enable seamless access to patients' complete medical history, providing context for current symptoms and improving prediction accuracy. The integration would involve secure API connections to national health infrastructure, enabling the system to retrieve and update patient records across the healthcare ecosystem. This would support continuity of care, reduce data duplication, enable longitudinal health tracking, and facilitate coordinated care across different healthcare facilities.

Mobile Application Development: Development of a mobile application would enable remote health screening and report access from smartphones, extending the system's reach beyond physical kiosk locations. The mobile app would provide the same functionality as the kiosk interface, with the added convenience of accessing healthcare services from anywhere. The app would support the full consultation workflow including registration, symptom entry, disease prediction, SHAP explanation viewing, and health report access, with push notifications for health updates and follow-up recommendations.

Telemedicine Functionality Integration: Incorporation of telemedicine functionality would allow patients to consult doctors through video or audio calls directly from the healthcare kiosk or mobile application. The telemedicine integration would involve scheduling and secure video

conferencing capabilities, with seamless transfer of prediction results and patient data to the consulting physician. This integration would create a comprehensive healthcare delivery platform combining AI-powered triage with human expertise, ensuring that patients receive appropriate care regardless of their location.

Enhanced Disease Prediction with Larger Datasets: Enhancement of disease prediction through training the model on larger and more diverse medical datasets would improve accuracy and reliability across different patient populations. This would involve continuous data collection and model retraining, with systematic evaluation of model performance. Federated learning approaches could be employed to train models on distributed healthcare data while maintaining patient privacy, ensuring the model remains current with evolving medical knowledge.

Cloud-Based Storage and Data Sharing: Implementation of cloud-based storage would securely maintain patient records and facilitate data sharing among authorized healthcare providers. The cloud storage would provide scalability, disaster recovery, and accessibility from any authorized location, with robust security measures including encryption and fine-grained access controls. Data sharing would support coordinated care and reduce redundant testing across different healthcare facilities.

Medical Image Analysis Support: Addition of support for medical image analysis using deep learning models would enable advanced diagnosis from X-rays, CT scans, ultrasound images, and skin lesion photographs. The image analysis would use convolutional neural networks trained on large medical image datasets, providing detection and classification of visual abnormalities. This feature would be particularly valuable in resource-limited settings without access to specialist radiologists.

Enhanced Data Security and Privacy: Strengthening data security through implementation of advanced encryption standards, role-based access control, and multi-factor authentication would ensure patient data remains protected from unauthorized access. Regular security audits, penetration testing, and compliance with healthcare data protection regulations would maintain the system's security posture and build patient trust.

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